

IMPERIAL COLLEGE LONDON

MASTER OF SCIENCE THESIS

DEPARTMENT OF AERONAUTICS

Experimental Study of Paraglider Aerodynamics

Author:

Sarah BECKER

Supervisor:

Dr. Paul BRUCE

*A thesis submitted in fulfillment of the requirements for the degree of MSc. in Advanced
Aeronautical Engineering in the Department of Aeronautics*

September 15, 2017

MASTER OF SCIENCE THESIS

DEPARTMENT OF AERONAUTICS

Experimental Study of Paraglider Aerodynamics

by Sarah Becker

Abstract

This project report presents experimental results that were obtained by running wind tunnel tests on a reduced-scale semi-rigid model of a ram-air wing, emulating two-dimensional behaviour. Ram-air parachutes are currently being increasingly studied as they are starting to be used in energy production, in addition to their numerous applications in the recreational field, notably sports, or military field and cargo delivery. This report experimentally investigates the role of the leading edge intakes configuration on drag, as well as the influence of the trailing edge configuration on aerodynamic performance. This report aims at producing experimental data useful in optimising the aerodynamics of ram-air parachutes, which can be used from a scientific point of view as well as directly by ram-air parachutes manufacturers.

Acknowledgements

I would like to thank Dr. Paul Bruce for all his assistance throughout the course of this project. His insights and experience in experimental projects have been of great help.

This project would not have been possible without the precious help of Mr. Roland and Mr. James, who greatly advised me and provided support in the manufacturing of the model and the experimental setting.

My thanks to Dr. Gouder and Mr. Dadd for their precious insights and their help.

Finally, I would like to extend my gratitude to my fellow coursemates who have assisted me and provided a motivating working environment over the course of this project.

Contents

Abstract	i
Acknowledgements	ii
1 Introduction	1
2 Literature review and background	2
2.1 State of the art on ram-air inflated parachutes	2
2.1.1 General description	2
2.1.2 Usual geometric and aerodynamic parameters	3
2.2 Review of key numerical and experimental studies on ram-air parachutes . .	5
2.2.1 Experimental studies	5
2.2.2 Properties of the flow around ram-air wings	6
2.2.3 Leads towards an optimised aerodynamic performance	7
3 Project objectives and methodology	8
3.1 Optimisation of paraglider performance	8
3.2 Methodology	9
3.3 Design of inlet configurations	10
4 Experimental model design	11
4.1 Model dimensions and speed selection	11
4.2 Airfoil selection	12
4.3 Ribs geometry and spacing	13
4.4 Cloth selection	14
4.5 Fabric cloth design	14
4.5.1 Pressure difference and inflation	14
4.5.2 Extra length of cloth between ribs	15
4.5.3 Stresses in the cloth	15
5 Wind Tunnel Testing	16
5.1 Estimation of performance	16
5.1.1 Parasitic drag estimation	16
5.1.2 Drag estimation	17
5.2 Methodology	18
5.2.1 Measurement methodology	18
5.2.2 Force balance settings and measurement precision	18
5.2.3 Definition of stall	19
5.2.4 Angle of attack correction	19
5.2.5 Blockage effect	20

6	Results	20
6.1	Force balance calibration	20
6.2	Base configuration	21
6.2.1	Side-inflated configuration	22
6.2.2	Improvement of base configuration: one-intake configuration	24
6.3	Influence of trailing edge deformation	26
6.4	Full-range inlet configuration	27
6.4.1	Initial configuration and improvement	28
6.4.2	Results of the full-range configuration	29
6.5	Optimised inlet and improved configuration	31
7	Discussion	33
7.1	Sources of imprecisions	33
7.2	Overall performance compared to previous results	34
7.3	Impact of trailing edge deformation as a drag mechanism	35
7.4	Impact of inlet size on drag	36
7.5	One-intake configuration	37
8	Conclusions and recommendations	38
A	NASA LSA-0417 airfoil coordinates	41
B	Inlet design - XFOIL study	42
C	Influence of Reynolds number on flow behaviour around NS(1)-0417 air- foil	43
D	Model drawings	44
E	Determination of cloth fabric Young modulus	45
F	Drag coefficients for parasitic drag (XFOIL)	46

List of Figures

1	Average Dutch wind speed with altitude (adapted from [1])	1
2	General description of a parachute [2]	3
3	Angle of incidence for stall of different ram-air parachutes compared to theory (solid line). Figure taken from [2]	4
4	Values of drag coefficient obtained from experiments on flexible (triangles) and rigid (diamonds) ram-air wings, compared to theory (solid line). Figure taken from [2]	5
5	Flow around an airfoil with (a) a round leading edge and (b) a leading edge cut. From [3]	6
6	Three-dimensional effect on flow around a ram-air wing at (a) 25% and (b) 50% of cell span. From [3]	7
7	Drag contributions for (a) a 36 m^2 glider and (b) a 300 m^2 glider. Figure taken from [2]	8
8	View of the rigid internal structure	9
9	Inlet configurations: (a) full range inlet and (b) optimised inlet	11
10	Schematic view of the model	11
11	Typical aerofoil sections historically used for ram-air parachutes	12
12	Sharknose [®] airfoil section	13
13	Rib used for the model	13
14	Model of a simply supported beam with uniformly distributed load	15
15	Shape of the inflated cloth, computed from equation 9	15
16	Drag coefficient of smooth cylinder [4]	17
17	Inclination of model with wind tunnel operating (b) compared to reference at 0° angle and at rest (a)	19
18	Force balance calibration	21
19	Parasitic forces and pitching moment	21
20	Under-inflation seen at (a) the leading edge and (b) the trailing edge	22
21	Performance of the side-inflation configuration	23
22	One-intake configuration: (a) front; (b) trailing edge	24
23	Performance of one-intake configuration (blue triangles); all-open cells (red diamonds); and side-inflation configuration (green squares)	25
24	Open-cells configuration	26
25	All-opened cells configuration trailing edge: (a) free; (b) with tape; (c) stitched with threads and middle free cell	27
26	Performance of all-opened intakes configuration with free leading edge (blue circles), taped leading edge (green triangles) and stitched trailing edge (red diamonds)	28
27	Full-range configuration	29
28	Ram-air parachute inflated leading edge. Adapted from Mpora website	29
29	Full-range intake configuration results	30
30	Optimised intake configurations	31

31	Performance of all inlet configurations compared to base configuration (black squares)	32
32	Comparison between modified "full-range" configuration results and values obtained on a similar semi-rigid model [5]	35
33	Comparison of lift-to-drag ratio obtained for different inlet sizes: (a) numerical results from Ross [6]; (b) experimental results from this project (black: one-intake configuration; blue: modified optimised configuration 5% inlet size; red: modified "full-range" configuration 7.5% inlet size	37
34	Flow around NASA LS1-0417 airfoil at $Re = 5 \times 10^5$ and at (a) -4° and (b) 15° . Computed with XFOIL	42
35	Influence of Reynolds number on stagnation range	42
36	Performance of NASA LS1-0417 at $Re = 5 \times 10^5$ (blue diamonds) and at $Re = 1.5 \times 10^6$ (grey circles)	43
37	Model drawings seen from: (a) front; (b) right; (c) up	44
38	Tensile test results on tested samples	45
39	XFOIL results for struts drag coefficients	46

List of Tables

1	Parasitic drag contributions	17
2	Results for all inlet configurations at 8° angle of incidence	33
3	Project modified "full-range" configuration results compared to previous literature results	36
4	NASA LS1-0417 airfoil coordinates	41
5	Coordinates obtained from XFOIL for stagnation points	42

1 Introduction

A total of 430 GW of power, which corresponds to approximately 7% of the world total generated power, was produced by wind power in 2015 [7], mainly by wind turbines. In the past 10 years, there has been a new trend in wind power: harvesting the energy of high-altitude winds [1]. At 1000 m altitude, winds are about three-times faster than surface winds as figure 1 suggests, going from about 3 m/s in average to 9 m/s. The equation between power and speed is as follows:

$$P = \frac{1}{2} \rho A V^3 \quad (1)$$

where A is the surface exposed and V the speed, meaning that for a three-fold increase in speed, the increase in power would be 27-fold for an equivalent surface. Furthermore, high winds are steadier and available almost all over the world, unlike surface winds which are intermittent and only exploitable in certain areas of the globe. Speed, steadiness and availability make harvesting high-altitude energy a very promising solution for generating clean power.

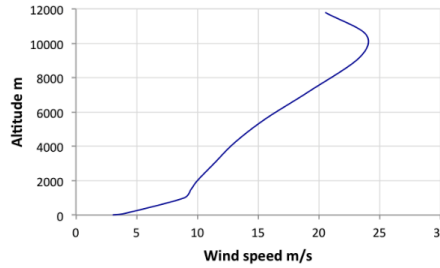


Figure 1: Average Dutch wind speed with altitude (adapted from [1])

One of the possibilities to exploit high-altitude wind power is to deploy kites tethered to the ground. As the kite flies, it uncoils lines from a drum on the ground that rotates and produces energy. To produce the equivalent of a 3 MW turbine, with a three-fold increase in wind speed, it would therefore be enough to deploy a 70 m² kite at 1000 m altitude [1]. The advantages of that technology are manifolds: kites are much cheaper and much lighter than wind turbines, and they can be deployed almost all over the world. However, this technology is still in its early stages, with much improvement to make before it can be used on an industrial scale.

Kite speed is an essential parameter in the successful development of this technology. Understanding the aerodynamics of kites (or ram-air parachutes) and improving their aerodynamic performance is thus key to optimising the energetic yield of such systems. Because of the proprietary nature of works on ram-air parachutes, a large portion of the research has not been made public. This project aims at contributing to the study of kites aerodynamics from an experimental perspective, focusing particularly on the influence of the leading edge air intakes on aerodynamic performance, and at providing experimental data on the performances of different inlet configurations. Previous experimental works have

vastly resorted to fully-flexible full or reduced-scale models [8] [9] [10], or rigid reduced-scale models [11]. Due to the great complexity of ram-air parachute aerodynamics, it has proven very hard to decouple the numerous drag mechanisms, resulting in an increased effort towards the development of numerical fluid-structure interactions models to better understand the aerodynamics and to numerically simulate ram-air parachutes behaviour. This study aims at providing experimental evidence of theories inspired by numerical works produced by Ross [6] and Belloc et al. [12]. In order to decouple the drag mechanisms, a semi-rigid scaled model, with a flexible fabric spread over a rigid internal structure, was used for this project. This project's objectives can be summed up as follows:

- Review the main experimental and numerical works on ram-air parachutes.
- Design and manufacture a properly scaled model which is representative of full-scaled ram-air parachutes behaviour.
- Study, from wind-tunnel testing, the contribution of the leading edge inlets on the total drag by comparing different inlet configurations to a sealed configuration; provide experimental evidence suggesting ram-air parachutes can be optimised with specifically designed inlets.
- Confront the experimental results with previous literature and outline the novelty of this experimental work.

2 Literature review and background

2.1 State of the art on ram-air inflated parachutes

2.1.1 General description

According to [2], ram-air parachutes were invented in the early 1960s. A ram-air parachute looks like a low-aspect ratio wing once it is inflated, as shown in figure 2. It is made out entirely of fabric, with no rigid parts, which guarantees that it can be packed and deployed similarly to conventional round-canopy parachutes. It consists of an ellipsoidal planform - historically rectangular, as in figure 2, but it has evolved towards an ellipsoidal shape - and an airfoil cross-section, which is maintained by airfoil-shaped ribs stitched chord-wise to the upper and lower membranes and disposed at regular span-wise intervals. Suspension lines are usually attached to every other rib to reduce drag. They are used to steer the parachute. A cell is demarcated by two loaded ribs.

The leading edge is usually open across the whole span to allow the inflation mechanism to take place. By this means, the air inside the canopy reaches stagnation pressure and the positive outwards difference of pressure results in the canopy maintaining its inflated shape.

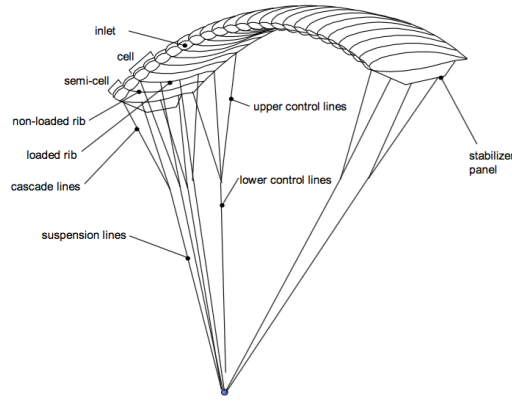


Figure 2: General description of a parachute [2]

2.1.2 Usual geometric and aerodynamic parameters

This section aims at describing the usual aerodynamic parameters for kites. These usual parameters will be used as references for this project model.

Aspect Ratio

Aspect ratio is a geometric parameter which greatly influences induced drag. It is defined as the ratio between the wing span and the projected wing surface.

E. Puskas describes the compromise that has been reached regarding aspect ratio for gliders in [13]. Basically aspect ratio up to 4 have been reached for conventional ram-air parachutes, but the practical upper limit is about 3. This is explained by the fact that any gain in drag reduction from a higher aspect ratio over that practical limit will be offset by the need for more and longer suspension lines. With recent progress, and more sophisticated configurations, aspect ratio can reach up to 4. Therefore, aspect ratio of 3 or 4 are usually seen for conventional ram-air parachutes.

Reynolds number

The Reynolds number Re greatly varies with each application. For paragliding as a sport, 1.5m to 2m chord gliders are used at speeds varying between approximately 3m/s and 10m/s. The range for the Reynolds number is given by the following equation:

$$Re_{sport} = \frac{U \times c}{\nu} \approx 3 \times 10^5 - 1.3 \times 10^6 \quad (2)$$

where the air viscosity at 1000 m is worth: $\nu = 1.58 \times 10^{-5} \text{ m}^2/\text{s}$.

For military and cargo applications, the chord can be much longer, reaching 10m, and the speed can easily reach 30-50m/s, which results in a much higher Reynolds number that can reach $Re = 3 \times 10^7$.

This project focuses mainly on high-altitude wind-powered energy- producing gliders. For these paragliders, chord is usually about 3m long, and the altitude and air speed they are designed for are respectively about 1000m and 8m/s, which gives the following target

Reynolds number:

$$Re_{energygliders} = 1.5 \times 10^6 \quad (3)$$

Angles of attack range

Small angles of attack are usually used in accelerating phases as the air penetration is better. Larger angles of attack are used in braking phases or during turns.

S. Lingard stated that ram-air canopies stall at lower angles of attack than rigid wings of corresponding aspect ratio and airfoil section [2]. As shown in figure 3, stall starts at approximately 9° angle of incidence for rigid (Notre Dame) and flexible (NASA) ram-air wings compared to theoretical data for ram-air rigid wings of corresponding aspect ratio of 3 and airfoil section (18% CLARK-Y airfoil). Ross suggests that this is probably caused by the sharp leading edge of the inlet upper lip which causes separation at relatively low lift coefficients [6].

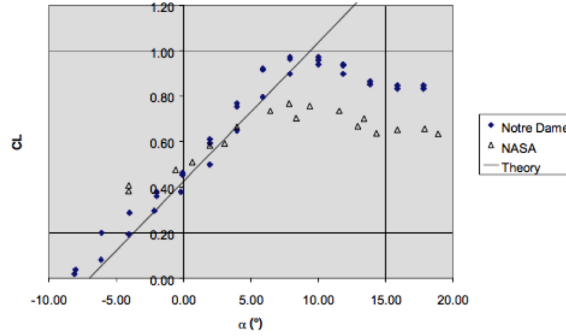


Figure 3: Angle of incidence for stall of different ram-air parachutes compared to theory (solid line). Figure taken from [2]

The range of angles of incidence that will be taken as reference in this project is therefore about -4° to stall (around $10^\circ - 15^\circ$). Angles of incidence lower than -4° are unlikely to be useful for high-altitude energy-producing paragliders as lift is negative at such angles of incidence.

Usual aerodynamic performance (C_l , C_d)

As seen in figure 3, the lift coefficient of traditional ram-air parachutes reaches about 0.8, which is far from what could be theoretically predicted (at least 1.2) from the airfoil section.

S. Lingard gives experimental data regarding the usual drag coefficient for conventional ram-air parachutes (same aspect ratio and airfoil section used as previously). The data can be seen in figure 4. The drag coefficient C_d for usual ram-air wings ranges from approximately 0.06 to 0.2 for the useful range of angles of attack. Note that all results in this section are for ram-air wings, without suspensions lines.

As a result the lift-to-drag ratio obtained for ram-air wings of aspect ratio of 3 reaches about 5 according to S. Lingard [2] and about 2 to 4 for ram-air parachutes (anhedral angle and suspensions lines taken into account) according to E. Puskas [13]. The maximum lift to drag ratio is reached at about $8 - 9^\circ$ angles of incidence (just before stall).

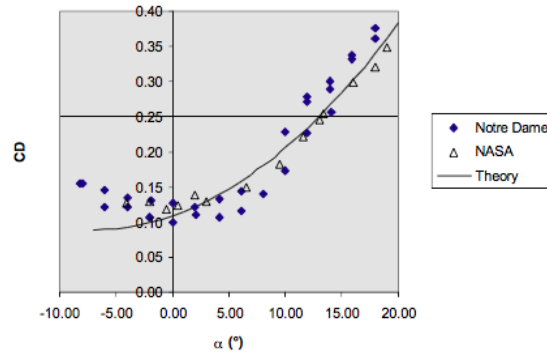


Figure 4: Values of drag coefficient obtained from experiments on flexible (triangles) and rigid (diamonds) ram-air wings, compared to theory (solid line). Figure taken from [2]

Suspension lines, anhedral angle

Incorporating a ram-air wing into a full parachute system requires the addition of suspension lines, usually connected to every other rib, and which constitute the means to steer the parachute. The addition of such lines obviously has a great impact on drag.

S. Lingard wrote that early designs opted for flat planforms, which resulted in very long lines and additional drag which clearly offset any gain in lift [2]. The optimum, empirically found, is to set all lines of any chordwise bank the same length, which gives the parachute an anhedral shape. The optimum ratio of line length to span is about 0.6 in order to get the higher lift to drag ratio. Lines and anhedral angles are not considered in this project however, as the focus is on leading edge influence on drag.

In this section, the main geometric and aerodynamic parameters for ram-air parachutes were reviewed. The key issues that have to be addressed to improve high-altitude wind-powered energy-producing kites performance are: control and stability of the automated trajectory; and aerodynamic performance optimisation (reducing drag while maintaining good inflation properties). This report focuses on the latter.

2.2 Review of key numerical and experimental studies on ram-air parachutes

2.2.1 Experimental studies

The aerodynamics of ram-air parachutes are very complex as they involve three-dimensional unsteady and turbulent flows over flexible and deforming boundaries. There are complex fluid-structure interactions as the shape of the parachute influences the flow and reciprocally. Because of such complexity, ram-air parachute design and development has mostly been an empirical process since its invention in 1964.

As Fogell pointed out, aircraft drop-tests for testing and certification of ram-air parachutes have been extensively used and are still useful today as it is obvious that great confidence in the design is required for such safety-critical pieces of equipment [3]. However, taking measurements in flight is very hard if not impossible to achieve and possibilities of applying instrumentation beforehand are very limited as parachutes need to be deployed

from a folded configuration [10]. An alternative to drop-tests is wind tunnel testing, which provides a more controlled environment, whether for full-scale or scaled models. Such tests have usually produced data such as force measurements and influence of aspect ratio, angle of attack, and lines configuration on the aerodynamic performance. For example, Ware and Hassell [8] and Nicolaides [9] have investigated scaled flexible ram-air parachutes and produced experimental lift and drag coefficients that were used as basis for figures 3 and 4 from Lingard's work [2]. These past works are also used as references later in this report, in section 7. In these works, lift to drag ratios were found to be between 2 and 6.

However, experimental testings fail to produce, except on rare occasions, usable data on the flow around the ram-air wing and inside the intake. Computational studies have been led in recent years to overcome the limitations of experimental works.

2.2.2 Properties of the flow around ram-air wings

The flow around a ram-air wing greatly differs from the flow around the corresponding airfoil. Fogell presents the impact of a leading edge cut on the flow around an aerofoil at $Re = 2.1 \times 10^6$ and at a 5° angle [3]. The flow around the pristine aerofoil, shown in figure 5a, is well-behaved and attached. However, the introduction of a leading edge cut greatly affects the flow, particularly on the bottom surface (see figure 5b). The flow is able to enter the internal space but is directed downwards towards the leading edge by a superior internal pressure. This causes flow separation at the bottom leading edge lip and results in the formation of a bubble of recirculating air on the bottom surface. As for the lower surface, the peak of low pressure is still present, though greatly reduced and detachment occurs towards the trailing edge. This is accompanied by a 15% decrease in lift and a 210% increase in drag compared to the original airfoil [3].

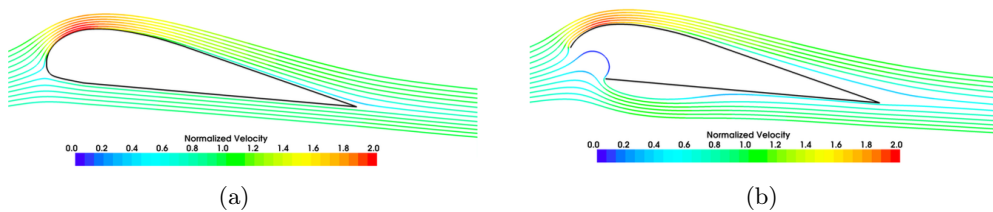


Figure 5: Flow around an airfoil with (a) a round leading edge and (b) a leading edge cut. From [3]

Flow perturbation by the introduction of a leading edge cut is further seen when studying three-dimensional effects of ram-air wing shape. Note, in figure 6, that the parafoil shape changes along the span compared to the original aerofoil. The inflation process greatly affects the aerofoil shape, as there is an overall increase in thickness, particularly noticeable at the trailing edge. The scalloping of the cell due to inflation results in an increase in cell height and therefore a larger obstruction to the flow. Compared to the 2D behaviour, the flow around the upper surface is greatly affected by the leading edge cut, as detachment occurs right after the peak of low pressure.

The flow is able to enter the internal space of the cell, like in the 2D behaviour, before

being forced to exit towards the leading edge bottom lip by higher internal pressure. The bottom surface separation bubble is larger and occupies about 60 – 70% of the bottom surface. This phenomenon is even worsened at half-cell. Overall, this results in a dramatic drop of aerodynamic performance: flow detachment causes the lift coefficient to be divided by over 3.5 compared to the cut aerofoil performance. Drag is increased by a further 2.5 factor.

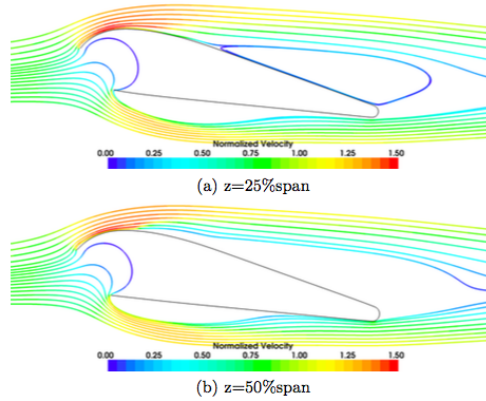


Figure 6: Three-dimensional effect on flow around a ram-air wing at (a) 25% and (b) 50% of cell span. From [3]

2.2.3 Leads towards an optimised aerodynamic performance

The drag-generating mechanisms in ram-air parachutes are manifold and frequently coupled with each other. The leading edge inlets have been found to be the greatest sources of drag, as they cause flow perturbation and flow detachment on both upper and lower surfaces of the wing [3] [6] [12]. The inlet shape and position also affects the inflation of the wing and thus the shape of the wing. Inflation results in spanwise scalloping of the canopy and in the creation of spanwise forces which draw the ribs inwards and modify the 3D shape of the wing. This causes the streamlines to spread out after reattachment and the flow to move from the central part of the cell to the ribs [3]. The inflated shape of the airfoil also increases pressure drag as the airfoil presents a larger obstruction to the flow. Other drag mechanisms include lines drag, wing tip effects, fabric porosity [5], trailing edge deflection (due to line tension). Ross and Fogell both suggest that the rounding of the trailing edge, seen in figure 6, might have a important impact on drag as it can cause flow separation in that region. Lingard summarises his view on the significance of the main drag mechanisms, presented in figure 7. Independently of the ram-air parachute size, the leading edge inlet is responsible for over 35% of the total drag [2]. Studies, mainly based on numerical simulations, have recently investigated the influence of the inlet position and size on the aerodynamic performance of paragliders. Ross suggests that the inlet size could be dramatically reduced (by more than half) compared to standard sizes without significant detriment to the performance in terms of angle of attack range but with significant aerodynamic improvement [6]: he has found that the lift to drag ratio for the 2D airfoil goes to about 30 to approximately 90 when the inlet size is reduced

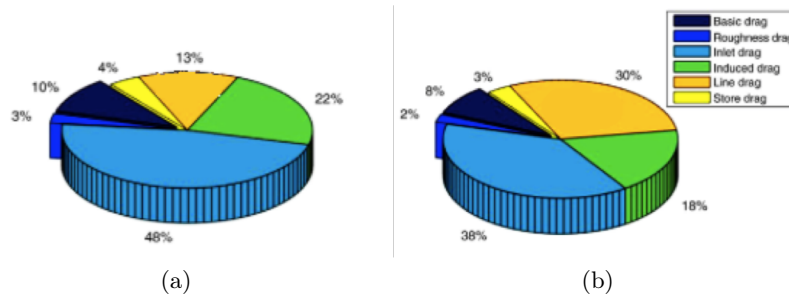


Figure 7: Drag contributions for (a) a 36 m^2 glider and (b) a 300 m^2 glider. Figure taken from [2]

from 8.4% to 4% of the chord. Belloc et al. have studied the decrease in aerodynamic performance on a 2D airfoil as the width of the inlet is increased [12]. They have shown that this decrease in performance might be slightly offset by positioning the inlet lower on the leading edge. Experimental studies have not been able to produce data similar to what numerical simulations have suggested. Ross provided experimental results showing that the lift to drag ratio of flexible parafoils went from 4 to 5 without noticing major inflation difference when reducing the inlet size [6]. Experiments on rigid models fail to produce the same results, as the role of the inlet is probably not properly captured. This project aims at providing further experimental evidence of the optimisation process of the inlet.

3 Project objectives and methodology

This project aims at experimentally studying the aerodynamics of paragliders, with a particular focus on identifying the influence of the inlet size on the aerodynamic performance.

3.1 Optimisation of paraglider performance

According to aforementioned studies and figure 7, inlet drag accounts for a major part of the total drag generated by a paraglider, between 35% and 50%. In order to improve the aerodynamic performance of high-altitude energy-producing paragliders, developing a better-performing intake therefore appears as the most obvious choice. As different computational studies suggest [6] [12], it should be possible to reduce the size of the air intake compared to what is being used currently, i.e. approximately 8% to 10% of the chord. This has given great results in numerical simulations: drag is greatly reduced and lift unaffected or slightly increased, resulting in a much better lift-to-drag ratio [6]. The aim of this project is to experimentally identify the influence of intake size on paraglider aerodynamics by comparing three different intake configurations, detailed further in subsection 3.3. In addition, the role of the trailing edge deformation (due to inflation) is to be identified as well and compared to the influence of the air intake on drag.

3.2 Methodology

This project aims at providing an experimental validation of an optimisation process that was obtained by computational simulations. The model used for this experimental project, further described in section 4, is a semi-rigid scaled model, constituted of a rigid internal structure (see figure 8) on which the flexible sail is spread. Three-dimensional effects were suppressed (or greatly reduced) by using a flat rigid internal structure.

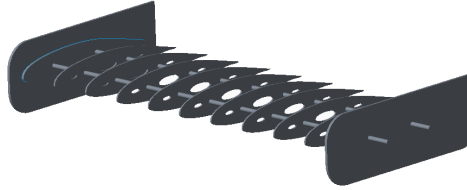


Figure 8: View of the rigid internal structure

The aim is to isolate the influence of the intake size and position on the aerodynamic performance and suppress as much as possible the influence of other drag mechanisms. A "2D" semi-rigid model facilitates force measurements compared to a fully flexible kite model as struts can easily be connected to the rigid parts. Likewise, the angle of attack is much more easily controlled and this model is free from lines as it is held by struts on the sides, thus eliminating the interference from line drag. Furthermore, with a fully-flexible reduced-scale paraglider model, all drag mechanisms are coupled: the deformation of the leading edge and the inflation process cause the ribs to be drawn inwards and result in cell deformation as well as canopy scalloping. Furthermore, changing the angle of attack requires a force to be applied on the lines and results in an undesired deflection of the trailing edge, which further adds to drag. It would be very hard among all those mechanisms to isolate either the intake influence or the trailing edge deformation influence on aerodynamic performance.

On the other hand, it is much easier to decouple all those effects on a semi-rigid model as the internal structure is fixed. Of course, despite the end-plates which limit wing-tip influence and emulate 2D-behaviour, there will be some 3D effects as the sail will scallop in between ribs. But these are indissociable from leading edge deformation and kite inflation. To get rid of all canopy deformation effects, a fully rigid model with a leading edge cut could be used, as K. Bergeron et al. did [11]. However, the main limitation of such a model is that it does not properly capture the intake behaviour, by not allowing any flexibility. For full-scale flexible parachutes, the inlet shape changes with inflation as the inlet size increases and the upper and bottom lips slightly scallop. The 2D semi-rigid scaled model is to be tested in a $1.37\text{ m} \times 1.22\text{ m}$ wind-tunnel section. The following sets of testing are to be carried out:

- Closed leading-edge configuration: the aim of that set of measurements is to get an overview of the performance of the paraglider model without any cut in the leading edge. The inflation will be done by an external system through the side of the model.

- "Full-range configuration": the sail cloth will be cut as shown in figure 9a along the leading edge of the model. This intake is 7.5% of the chord in length.
- Optimised configuration (see figure 9b): this intake is smaller and located on the lower surface. This configuration is to be compared to the previous two and should perform better than the 7.5% configuration.
- Constrained trailing edge configuration: in that set of testing, the trailing edge is not allowed to deform upon the last 15% of chord on both the upper and lower surfaces. This will be compared to the previous configuration with a free leading edge.

The design of both intake configurations is explained in section 3.3.

3.3 Design of inlet configurations

The aim of the inlet is to allow paraglider inflation through ram-air pressure. One of the aims is to maximise internal pressure inside the kite to ensure that the shape is retained through all possible flight uses. In order to maximise internal pressure, the intake should be located at the stagnation point, where pressure is at its peak. However, the stagnation point position depends on the angle of attack.

Using XFOIL, the whole stagnation point range was computed for the chosen airfoil, NASA LS(1)-0417 (see subsection 4.2 for airfoil selection) and the chosen Reynolds number, 5×10^5 (see subsection 4.1 for Reynolds number selection) and the chosen range of angles of attack, -4° to 15° .

There is a compromise to be reached when designing a paraglider inlet: if the intake is located near the front of the stagnation pressure range, excellent internal pressure will be reached at low angles but the performance at higher angles will be very poor, with parachute stall reached at low angles; if the intake is located towards the end of the stagnation range, there is likely to be a frontal collapse of ram-air wing nose at low angles of attack, which results in poor inflation and is detrimental to performance.

Two intake configurations are studied in this project. The first one, referred to as "full-range inlet" and pictured in figure 9a, covers the whole range of angles of attack computed with XFOIL. The cut is oriented at 48.1° with respect to the chord, the length represents 7.5% of the chord and the vertical height is 3.5% of the chord. This configuration is expected to perform correctly at all angles of attack. The second configuration, referred to as "optimised inlet" and pictured in figure 9b, is based on Ross's description of the optimised inlet [6]. It covers the stagnation range which gives lift coefficients from 0.5 to 1.5 for the airfoil at the chosen Reynolds number. The cut is oriented at 41.8° with respect to the chord, its length represents 4.9% of the chord and its height 3% of the chord. This configuration is expected to lead to a much better aerodynamic performance with the inlet being 36% smaller than in the full-range configuration. However the behaviour at high angles of attack is unknown, as the stagnation point is likely to be out of the inlet. The aim is to identify the importance of the reduction in intake size on the aerodynamic performance, as well as verifying that such an intake configuration could be used by paraglider manufac-

turers - meaning that stall is not reached at low angles of attack and that it stays inflated at low angles of attack. XFOIL details for the inlets design can be found in appendix B.



Figure 9: Inlet configurations: (a) full range inlet and (b) optimised inlet

Note that a comparison between the stagnation range for $Re = 5 \times 10^5$, $Re = 7.5 \times 10^5$ and $Re = 1.5 \times 10^6$ can be found in appendix B. This concludes that the Reynolds number only has a very slight influence on the range.

4 Experimental model design

4.1 Model dimensions and speed selection

The wind tunnel used for the experiments is 1.22m high and 1.37m wide. The model consists of a flexible cloth glued to a rigid internal structure: eleven rigid ribs at regular intervals mounted on two 8mm diameter rods. Two end-plates are employed at the ends of the rods to eliminate wing-tip effects and simulate 2D behaviour. The model is then mounted on a force balance by the means of: two front struts, which are connected to both ends of the front rod with bearings; and a back strut, solidly connected to the model back rod. Figure 10 shows the rigid structure with the ribs, front rods, end plates and the rear part which is connected to the back strut. The back rod is approximately 480 mm (which is equal to 1.6 times the chord length) downstream of the model to prevent any flow interactions between the wake and the back rod. In order to be able to vary the

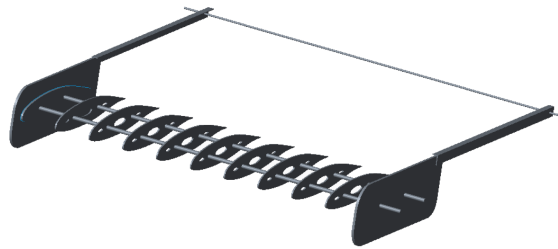


Figure 10: Schematic view of the model

angle of attack, the model can rotate around the front rod when the back strut is moved forwards or backwards by means of an extendable jack. With both front mountings and back mountings, the force balance gives the lift, drag, and pitching moment.

Model dimensions and aspect ratio

In order to reproduce 2D behaviour, the model was made as long as possible in the spanwise direction: the span reaches 755 mm which is the maximum allowed by the force balance

front struts. The chord was determined by taking into account the Reynolds number and the blockage effect (see section 5.2.5). It was set at 300 mm, which means the aspect ratio is 2.52. Ideally, to simulate 2D behaviour, the aspect ratio should be as high as possible, since 2D behaviour corresponds to an infinite span and therefore an infinite aspect ratio. However, the span is limited by the maximum spacing between the front struts. The other way to achieve a higher aspect ratio would be to decrease the chord. However, this would reduce the Reynolds number which was set to be able to emulate the same aerodynamic flow behaviour as for full-scale ram-air parachutes. Thus the aspect ratio of 2.52 was considered to be an appropriate compromise. Limited 3D effects are expected to happen but will be negligible compared to the effect of leading edge inlet. Furthermore, this aspect ratio matches usual ram-air wings aspect ratio as discussed in section 2.1.2, based on E. Puskas's works [13].

Reynolds number

The wind tunnel speed used for the experiment is 25.2 m/s (the upper limit for the wind tunnel used is about 30 m/s), which means the Reynolds number is:

$$Re_{exp} = \frac{U \times c}{\nu} = 5.0 \times 10^5 \quad (4)$$

which is to be compared to the Reynolds number of ram-air parachutes used to exploit wind-power, shown in section 2.1.2 to be around 1.5×10^6 , i.e. three times the experimental Reynolds number. However, the data obtained for lift, drag and pitching moment coefficients from XFOIL for the airfoil section used for the model (see section 4.2 for the airfoil selection process), shows that the flow behaviour at $Re = 5.0 \times 10^5$ is very similar to the results for $Re = 1.5 \times 10^6$. The comparison between both is presented in appendix C.

4.2 Airfoil selection

Several airfoils have traditionally been used for ram-air parachutes, according to numerous sources, including [2]. Different versions of the CLARK-Y airfoil (see figure 11a) have been extensively experimented but recent trends, motivated by research in glider technology, consist in using low-speed sections such as NASA LS(1)-0417, shown in figure 11b.

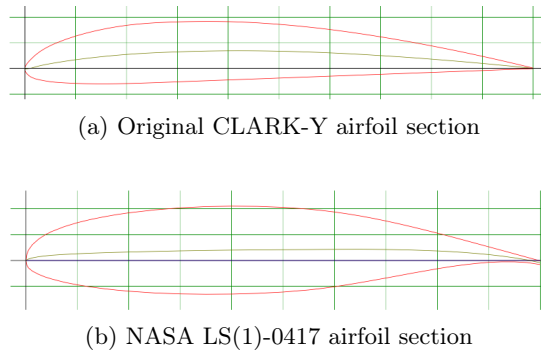


Figure 11: Typical aerofoil sections historically used for ram-air parachutes

Other airfoils are starting to be experimented, such as SharkNose[®] by Ozone (see figure 12), which is specifically designed for ram-air parachutes, but the lack of experimental or numerical data on such airfoils make them inappropriate for this project. In this project,



Figure 12: Sharknose[®] airfoil section

the objective is to produce experimental data which can be compared to other numerical or experimental evidence, in order to assess similarities or differences in the result. The best airfoils in that regard would be either one of the CLARK-Y airfoils or the NASA LS(1)-0417 which are traditionally used by kite manufacturers and for which extensive data is available. The NASA LS(1)-0417 was ultimately chosen for the following reasons:

- It is still being used by kite manufacturers whereas the CLARK Y airfoils seem to have fallen in disuse.
- Its aerodynamic performance at low Reynolds numbers is better than the Clark-Y airfoils as it generates less drag [2].
- It is thicker (18% compared to 12%) which means the ribs are stiffer (especially towards the trailing edge), which facilitates model manufacturing.
- Previous studies have used the NASA LS(1)-0417 airfoil to investigate the influence of inlet size on aerodynamic performance and thus offer available basis for comparison for this project experimental results [6] [12].

4.3 Ribs geometry and spacing

The ribs are laser cut out of 3 mm thick acrylic sheets. The thickness should be under 1 mm if it were to respect scaled dimensions, however thickness below 3 mm does not ensure sufficient rigidity of the ribs. As such, it was set at 3 mm. Openings were cut out in the ribs, as seen in figure 13, to ensure pressure equalisation between cells across the whole span. The spacing between each rib was set at 72 mm, yielding a total of 9 internal ribs



Figure 13: Rib used for the model

(and 2 external ribs) across the span. This gives a rib spacing over chord ratio of 24%. This ratio has been determined for a dozen different industrial (mainly sport) paragliders, using the average chord, and it ranges from 15% to 25%. The chosen ratio is within that range.

The rods are placed at 25% and 65% of the chord. The vast majority of lift is generated by the front end of the aerofoil, as shown in figure 6. Thus, almost of the lift is transmitted, through the front rod, to the front mountings of the force balance.

4.4 Cloth selection

The fabric used for the model is a sheet of woven polyester fabric (CPN180) made by Dimension Polyant and commonly used in the kite industry. Its thickness is approximately 0.2 mm, and the Young modulus 515 MPa. The Young modulus in the fibres direction was determined in appendix E, using tensile tests results provided by Dimension Polyant on fabric samples. The cloth is spread out such that the fiber direction is aligned with the span, thus giving stiffness in the spanwise direction.

According to the data found in [14], the minimum tensile strength of polyester fabric of about 0.2 mm thickness is approximately 500 MPa while the elongation at failure is about 20%.

4.5 Fabric cloth design

4.5.1 Pressure difference and inflation

As inflation occurs in the kite, the fabric deforms due to the difference between internal and external pressure. The pressure inside the kite is overall equal to the stagnation pressure while the external pressure varies alongside the chord.

From the definition of the pressure coefficient C_p :

$$C_p = \frac{p - p_0}{q_0} \quad (5)$$

where p is the static pressure at the point of interest; p_0 the free-stream static pressure; $q_0 = \frac{1}{2}\rho V_0^2$ the dynamic pressure; and V_0 the free-stream velocity; and from Bernouilli's equation:

$$p_0 + \frac{1}{2}\rho V_0^2 = p_0 + q_0 = p_{stagnation} \quad (6)$$

the following equation was obtained, for the difference between internal and external pressure Δp :

$$\Delta p = p - p_{stagnation} = p - p_0 - q_0 = q_0 \left(1 + \frac{p - p_0}{q_0} \right) = q_0 (C_p - 1) \quad (7)$$

The experimental dynamic pressure $q_{0_{scaled}}$ is about 8.5 times higher than the actual dynamic pressure, the ratio being given by:

$$\frac{q_{0_{scaled}}}{q_{0_{real}}} = \frac{\rho_{scaled} V_{0_{scaled}}^2}{\rho_{real} V_{0_{real}}^2} \approx 8.5 \quad (8)$$

where $\rho_{real} = 1.1 \times 10^{-5} \text{ m}^2/\text{s}$ at 1000 m altitude.

The XFOIL study of the airfoil showed that the difference in C_p between the actual Reynolds number (approximately 1,500,000) and the Reynolds number used in the ex-

periment (500,000) is negligible (see appendix C). This means that the pressure difference is much higher in the experiment. However, because the model is semi-rigid and not fully flexible, the pressure difference only impacts the fabric cloth, and has no effect on the ribs or on the model geometry otherwise. This means that the only difference that can be expected is a small difference in the cloth shape. The next section, section 4.5.2, explains how the cloth was designed to have a particular shape independently of the pressure difference, meaning that the scaled model will have the same behaviour as real ram-air parachutes, despite the difference in dynamic pressure.

4.5.2 Extra length of cloth between ribs

The spacing between two ribs is set at 72 mm. In order to incorporate fullness due to inflation in the design, extra cloth material was added between the ribs.

It was assumed that the cloth takes the shape of a catenary, as shown in figure 15. Indeed, in 2D, it can be considered as a simply supported beam subject to a uniformly distributed pressure load, as presented in figure 14. As such, the shape of the cloth in 2D is given by

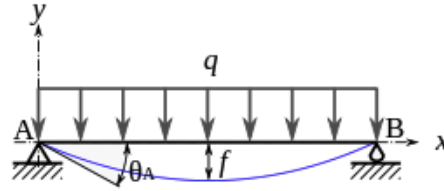


Figure 14: Model of a simply supported beam with uniformly distributed load

the following equation:

$$y(x) = \theta_A \times \frac{x}{L^3} (x^3 - 2Lx^2 + L^3) \quad (9)$$

where θ_A is the initial angle and L the length of the undeflected beam.

θ_A was estimated using information given by a kite manufacturer to be between 30 and 35° for such a rib spacing to chord ratio.

To get such an inflated shape, 5 mm of cloth length were added, setting the initial angle at approximately 32° and meaning that 77 mm of cloth are needed for each 72 mm rib spacing.



Figure 15: Shape of the inflated cloth, computed from equation 9

4.5.3 Stresses in the cloth

This section aims at verifying that the stresses in the cloth and the corresponding deformations do not exceed respectively the strength and the maximum elastic strain of the

material.

The cloth is a very thin membrane, simply supported on the ribs. It approximately takes the shape of a very thin half-cylinder between ribs. Even though that approximation is incorrect, as the radius of curvature actually varies along the cloth, this will give results of a similar magnitude. This means the stresses are given by the following equation:

$$\sigma = \frac{\Delta p \times r}{t} \quad (10)$$

where t is the thickness and r the radius of curvature of the cylinder, given by the following equation:

$$r(x) = \frac{(1 + y'(x)^2)^{\frac{3}{2}}}{y''(x)} = \frac{\left[1 + \left(\frac{\theta_A}{L^3} [4x^3 - 6Lx^2 + L^3]\right)^2\right]^{\frac{3}{2}}}{\frac{\theta_A}{L^3} (12x^2 - 12Lx)} \quad (11)$$

Close to the rib, where the stresses are the greatest, at $x = 0.5$ mm, the local radius of curvature is 2.29 m.

Taking the average C_p at -0.5, this gives $\Delta p = q \times (C_p - 1) = -562.5$ Pa. The corresponding tensile stress is then 6.5 MPa, which is clearly less than the tensile yield stress of the material. The corresponding strain is:

$$\epsilon = \frac{\sigma}{E} \approx 1.25\% \quad (12)$$

which is similarly far from the allowable strain.

5 Wind Tunnel Testing

5.1 Estimation of performance

5.1.1 Parasitic drag estimation

In this section, the interference drag estimations are shown. They are compared with the predicted performance of the model in section 5.1.2 to check that the setting is appropriate. The contributions to drag of all the set-up elements have been listed in table 1. The front struts are divided into three parts. The top part is covered by a NACA0022 airfoil and is referred to as part 1. The second part is a rectangular section which should have been covered by an airfoil section. However, during the set-up, due to a manufacturing defect, it was not possible to position the airfoil section. The third part connects the model to the struts and is streamlined with a NACA0016-airfoil-similar shape.

All elements are streamlined bodies, except the rod and the rectangular section (part 2) of the front struts. The end plates and corner sections are modeled as thin plates parallel to the flow as their aspect ratio (defined as the ratio of the width of the section to its height) are lower than 0.1. As such, the drag coefficient for those elements is obtained using equation 13, obtained from [4], valid for thin plates for Reynolds number between

Item	Qty	Shape	L (mm)	H (mm)	AR	Re	C_d	Drag (N)
Front struts 1	2	NACA0022	165	370	2.24	3×10^5	0.0123	0.58
Front struts 2	2	Blunt	22	100	3	4×10^4	1.3	2.22
Front struts 3	2	NACA0016	20	120	6	3×10^4	0.038	0.07
End plates	2	Streamlined	360	120	0.03	6×10^5	0.0052	0.17
Corner plates	2	Streamlined	650	20+12	< 0.1	10^6	0.0047	0.075
Back strut	1	NACA0016	20	875	44	3×10^4	0.038	0.252
Back rod	1	Blunt	4	700	175	6×10^3	1	1.03
Subtotal	-	-	-	-	-	-	-	4.40
Miscellaneous	-	-	-	-	-	-	-	0.22
Total	-	-	-	-	-	-	-	4.62

Table 1: Parasitic drag contributions

5×10^5 and 10^7 .

$$C_d = \frac{0.074}{Re^{0.2}} \quad (13)$$

The front struts drag coefficient was computed with XFOIL using a NACA0022 section at $Re = 2.6 \times 10^5$ and 0° angle of attack. The back strut and bottom part of the front struts drag coefficient is computed using XFOIL for a NACA0016 section at $Re = 3.2 \times 10^4$ and 0° angle of attack. Details of the XFOIL computation are provided in appendix F.

The back rod is a 4 mm diameter cylinder at low Reynolds number (approximately 6,400). Using figure 16 obtained from [4], the drag coefficient is taken at 1. The rectangular shaped part of the front struts has an aspect ratio of 3. According to Blevins, for a Reynolds number above 10^4 , this means that the drag coefficient of such a shape is 1.3 [15].

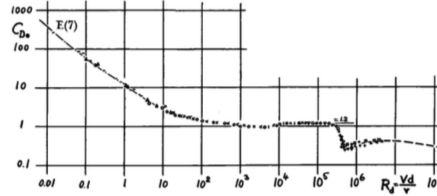


Figure 16: Drag coefficient of smooth cylinder [4]

After adding all those contributions, an additional 5% of the result is added to account for miscellaneous drag (bolts, nuts...). The total parasitic drag calculated is about 4.62 N. That is to be compared to the estimated drag of the kite.

Note that the rectangular section of the front struts was intended to be covered by an airfoil section (NACA0022). That would have brought the contribution of that part of the struts to be 0.15 N instead of 2.22 N. The total parasitic drag would have been 2.45 N, almost reduced by half.

5.1.2 Drag estimation

Experimental studies on fully flexible kites have shown that the drag coefficient of inflated kites is always greatly superior to that of the airfoil. According to numerical as well as experimental studies, it is at least above 0.065 [2] [5]. It can reach 0.3 for high angles of

attack. This would mean the amount of drag generated by the model is:

$$D = \frac{1}{2}\rho V^2 C_d S = 5.6 - 25.8 \text{ N} \quad (14)$$

This means the ratio of parasitic drag to kite drag is in the range given in the following equation:

$$0.18 < \frac{D_{\text{parasitic}}}{D_m} < 0.82 \quad (15)$$

With the intended set up (the front struts covered in their entirety), this ratio would have been between 0.1 and 0.43 at all times, which is significantly better. This ratio means that the parasitic drag accounts for between 20% and 80% of the total drag, which is not optimal, as it would be better to keep that ratio as low as possible. However, the parasitic drag was precisely measured before the experiments in section 5.1.1, in order to avoid errors and make sure that drag was properly measured.

5.2 Methodology

5.2.1 Measurement methodology

The model is placed in the wind tunnel section and connected to the wind balance front and back mountings via three struts. The wind tunnel is run at a speed of 25.2 m/s. The speed is measured by a Pitot tube set on the wind tunnel roof with a precision of ± 0.025 m/s. The model is originally set at a 0° angle of attack. The angle of attack is increased or decreased by displacing the rear strut by increments of 1° maximum. Increments of 0.5° have also been used to give a more detailed description of some configurations. Once the angle of attack is changed, measurements are taken for lift, drag and pitching moment using the force balance, which is further described in section 5.2.2. Each set of measurements was done twice to ensure robustness of the results. In the rare cases where the difference between the two values was greater than 5% of the total result, an additional measurement was made. The final result is the average between the two (or three) measured values. All results were normalised by the dynamic pressure q as follows:

$$\begin{cases} C_i = \frac{F_i}{\frac{1}{2}\rho b c V^2} \\ C_m = \frac{M}{\frac{1}{2}\rho b c^2 V^2} \end{cases} \quad (16)$$

where b is the span, c the chord, C_i the aerodynamic coefficient corresponding to the force F_i (lift or drag).

5.2.2 Force balance settings and measurement precision

Lift and drag forces as well as pitching moment are measured by a mechanical force balance in pounds (force) and foot-pounds, later converted to newtons and newton meters. The moment measured is the pitching moment around the first rod, at 25% of chord. Every

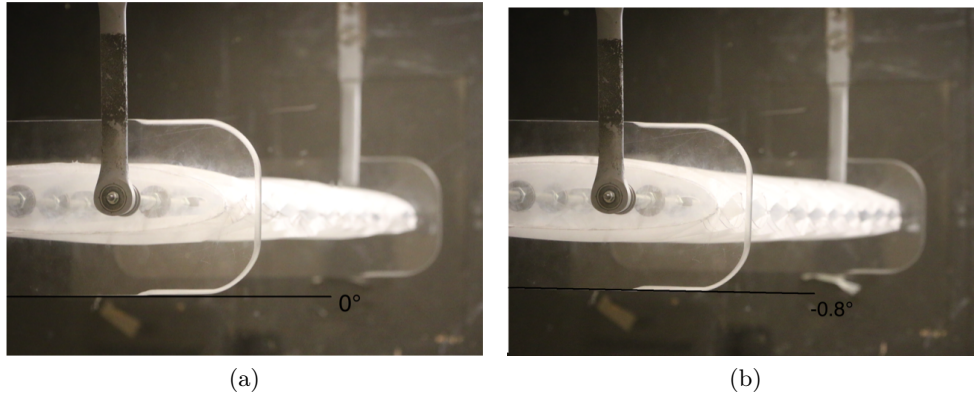


Figure 17: Inclination of model with wind tunnel operating (b) compared to reference at 0° angle and at rest (a)

measurement requires to balance the forces manually. This means that measurements are not obtained by time-averaging a set of values. The measurements can be read with the following precisions: ± 0.01 lbf (i.e ± 0.0448 N) for the lift; ± 0.001 lbf (i.e ± 0.00448 N) for the drag; and ± 0.001 ft·lbf (i.e ± 0.00136 Nm) for the pitching moment. Using the performance of the model obtained in the experiments, detailed in section 6, this means that the precisions for lift, drag and pitching moment are respectively in average 2%, 0.6% and 3% at 5° (the median value of angles of attack). The force balance has to be calibrated before every set of measurements.

5.2.3 Definition of stall

As mentioned above, measurements are obtained after balancing the forces manually, which means there is no possibility to do data acquisition over a period of time and then obtain a value time-averaging the data. When the ram-air wing starts to oscillate strongly when high angles of attack are reached, it is not possible to get an accurate measurement of the forces. Strong oscillations have been defined as stall: when the ram-air wing is no longer stable. This means that stall might not be clearly seen, in the figures of section 6, as a gradual decrease in lift. However it is considered that the large oscillations that make measurements impossible can actually satisfactorily characterise stall. The pitching moment was measured as well and change in slope can help identify stall.

5.2.4 Angle of attack correction

During the experimental testing, a significant difference was noted in the angle of attack when the wind tunnel is operating and when the model is at rest. Therefore, the angle read on screen, which is based on the position of the rear strut, is not the actual angle with wind speed. The 0° position was defined to correspond to a 0° inclination of the right-side end-plate bottom surface at rest, as shown in figure 17a. The angle reader was calibrated on that original position. The actual angle of attack was determined for each position using photographs and comparing them to the original 0° position photograph, as shown in figure 17.

5.2.5 Blockage effect

In a wind tunnel, the model under study is immersed in an air stream that is bound by the lateral walls of the tunnel. The flow past such a body is subject to blockage constraint: the presence of rigid lateral walls blocks the lateral displacement of the flow in the vicinity of the body, resulting in higher velocities than what would be found in an unbound airflow [16]. Particular attention has to be brought on proper scaling of the model dimensions in order to avoid undesirable blockage effects in the wind tunnel, which affect particularly the results for the drag coefficient. Numerous studies, among which Maskell's work [16], have suggested correction factors to be applied to the results. However, such correction factors depend on the body shape, as it differs for bluff and streamlined bodies, and no studies have been found regarding the correction factors to be applied to ram-air wings. Therefore, the blockage ratio was kept low in this project in order ensure the effect would be minimum and could be neglected. Choi and Kwon suggest that for avoiding the effects of excessive distortion of the flow, the allowable blockage ratio has to be under 5%, which is the upper limit used in numerical codes [17].

The angle of attack upper limit is 15° in this project. This means the blockage ratio is given by:

$$\frac{bc \sin(15^\circ)}{S} = 3.8\% \quad (17)$$

where b is the span, 0.8 m, c the chord, 0.3 m and S the wind tunnel area. When the struts, back rod and the projection of the corner plates are added, the total blockage ratio is: 4.9%. This explains why the data were used without applying a blockage effect correction factor.

6 Results

6.1 Force balance calibration

The first set of measurements is used to determine the parasitic drag generated by the end-plates, the struts and the back rod, estimated in section 5.1.1.

In order to determine the parasitic lift and drag forces, the experiment was run without the model. The end plates were connected to the front struts with a 8 mm diameter 800 mm long cylindric rod to give enough stiffness to the model. The end-plates holes were taped. The contribution of the cylindric rod was then deducted from the obtained results. This contribution was determined using $C_d = 1$ (see figure 16), which yields a drag of 2.489 N. Figure 18 shows the wind tunnel set-up for the force balance calibration.

The pitching moment has to be calibrated as well. When the angle of attack is changed, the back mounting is displaced, generating an additional parasitic moment which is independent of the aerodynamic pitching moment. In order to calibrate the pitching moment, the angle of attack was changed at rest with the full model on. The results for parasitic lift, drag and pitching moment are given in figure ??

The parasitic lift is negligible, it corresponds to less than 1% of the lift obtained for each angle of incidence. The small amount of lift obtained in the calibration measurements is

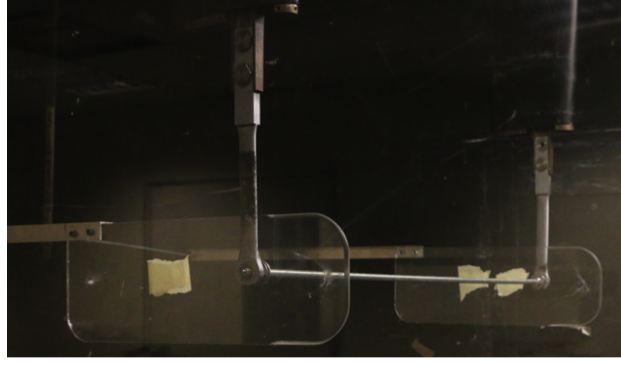


Figure 18: Force balance calibration

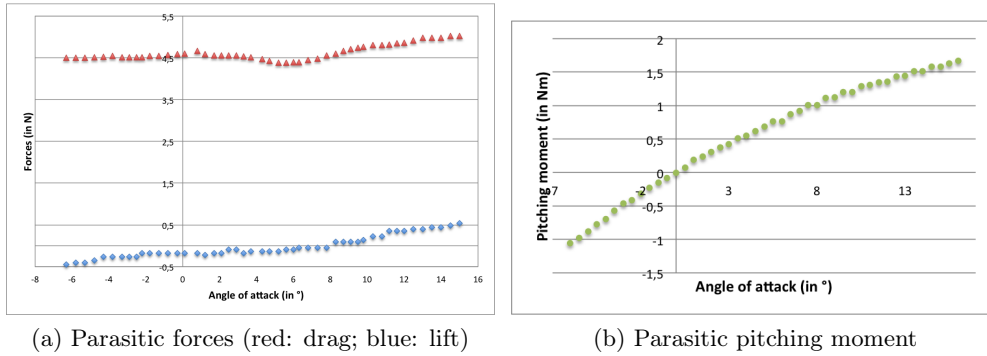


Figure 19: Parasitic forces and pitching moment

probably generated by the end-plates.

The parasitic drag that was found (once the cylindric rod drag was deducted) is close to what was expected from earlier estimations in section 5.1.2: about 4.5 N of drag were found up to 7°, with a slight increase afterwards up to 5 N. These results match the estimations which were calculated at 0° angle of incidence and gave 4.62 N of drag. The sharp increase in drag after 7°, matched by a similarly sharp increase in lift, probably corresponds to some flow detachment at the front part of the end-plate upper surfaces at higher angles of incidence.

The parasitic pitching moment was measured by varying the angle of attack of the whole model mounted on the force balance, with the wind tunnel at rest. The pitching moment varies approximately linearly with the angle of attack until 7°. The curve seems to flatten slightly thereafter. This is understandable as the angle of attack varies linearly with the back mountings displacement for low angles of attack. For slightly higher angles, smaller displacements are required for an equivalent angle change.

The results obtained in this section were deducted from the values obtained for every set of measurements in the experimental testing.

6.2 Base configuration

The aim of this section is to describe the results obtained for the base configuration, which is the inflated wing without intake. This configuration is then used as the basis of comparison for the different intakes.

6.2.1 Side-inflated configuration

The aim was initially to use ram-air pressure to inflate the model through an opening in the left-side end-plate. A tube was placed at the bottom of the wind tunnel parallel to the air flow. A funnel was mounted at the end of the tube. The other end was connected to the model left-hand side opening. This device ensures that the pressure at the entrance of the tube is equal to ram-air pressure.

The model is sealed at the leading edge and the trailing edge. Initially, it was hoped that only one piece of fabric could be used for the whole wing and then sealed at the trailing edge only. However, manufacturing issues resulted in two pieces of fabric being used - one for the upper and one for the bottom surfaces. The leading edge was sealed using tape, while the trailing edge was glued on a very thin layer. Despite the model being sealed, leakage was not totally prevented, and leakages combined with the slight pressure loss across the tube resulted in the model being slightly under-inflated, as seen in figure 20.

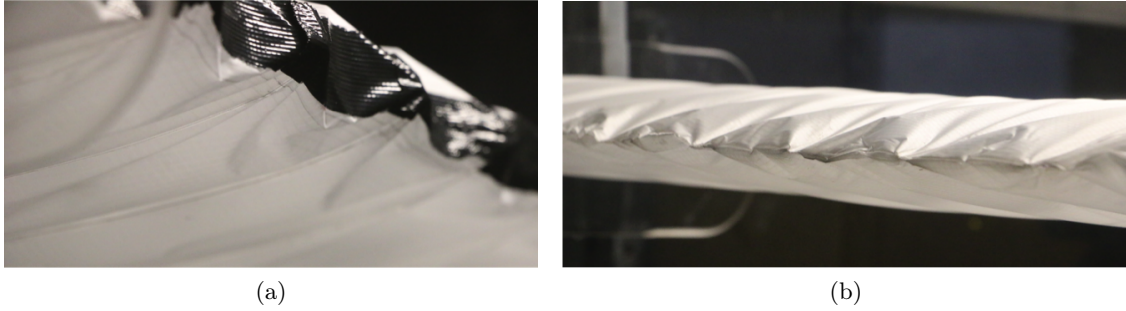


Figure 20: Under-inflation seen at (a) the leading edge and (b) the trailing edge

Figure 20a shows that there is a frontal collapse at the leading edge. The leading edge is not round but flattened inwards. This results from the equilibrium between external and internal pressure at the stagnation point. In this region, the external pressure is equal or even slightly higher (because of under-inflation) than the internal pressure, resulting in frontal collapse of the aerofoil shape. This causes decrease in lift and an increase in drag. Moreover, because of the pressure inside the model being inferior to the stagnation pressure, when the inclination starts to be noticeable (approximately $\pm 3^\circ$) and the stagnation point is displaced, the model deforms as the upper or lower surfaces (for respectively negative and positive angles of attack) are sucked inwards both at the leading edge and trailing edge. This phenomenon can be seen in figure 20a with ripples being present on the bottom surfaces. As the angle of attack is increased, these ripples gradually expand. These under-inflation problems are not resolved when taping the trailing edge to ensure less leakage. The results for this configuration are presented in figure 21. The additional amount of drag generated by the ram-air device at the bottom of the wind tunnel was measured without the model on and was found to be 0.51 N. This amount, in addition to the parasitic drag, was deducted from the drag forces measured. The ram-air device (tube and funnel) causes an increase in blockage ratio of the wind tunnel that is hard to quantify but is small compared to the model: therefore, no additional coefficient was applied to account for this slight increase in blockage ratio.

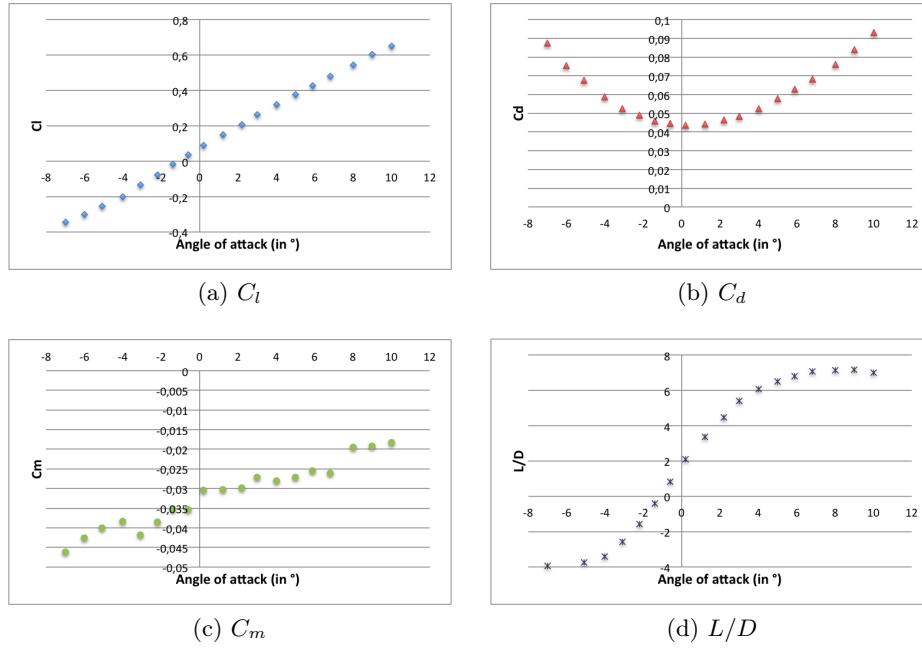


Figure 21: Performance of the side-inflation configuration

The lift coefficients obtained are similar to what is expected for a ram-air wing. Because of the frontal collapse of the leading edge, no additional amount of lift is generated compared to a ram-air wing. However, stall occurred at lower angles of attack (about 10°), due to wing under-inflation. Thus the lift coefficient could not reach more than 0.65, while it can go up to 0.8 for usual ram-air wings.

The drag coefficient C_d that was measured is lower than what is expected of a ram-air wing, despite the under-inflation, which is explained by the absence of intakes which greatly contribute to drag, as well as the absence of 3D deformations: for ram-air wings, changing the angle of attack usually requires deflecting the trailing edge, this deflection impacting the chord-wise shape of the wing and resulting in larger flow obstruction and therefore additional drag. Usually the minimum drag coefficient for ram-air wings is about 0.65 compared to 0.445 that was measured for this side-inflated configuration, which corresponds to a 30% decrease.

The model becomes unstable, as it starts to shake greatly, at 10° . These oscillations make it impossible to measure values for lift, drag and pitching moment. However, the lift coefficient is still linear up to that angle, as observed in figure 21a, without experiencing a decrease in slope as expected before stall. Likewise, the drag coefficient curve does not show a significant increase in slope, nor the pitching moment slope experience any change at 10.5° which would be further indicators of stall. This behaviour at 10.5° might be explained by the deformation undergone by the model as the angle of attack is increased: the bottom surface is sucked inwards, resulting in loss of lift and dramatic oscillations at much lower angles than expected. In conclusion, these results cannot be used as the base configuration to compare the rest of the results to, because the under-inflation results in a decrease in performance and very early failure. Another attempt was made to obtain results for a base configuration, described in the following section 6.2.2.

6.2.2 Improvement of base configuration: one-intake configuration

The problem with the previous configuration was the under-inflation resulting from the fact that it was not possible to achieve high enough pressure inside the model. In order to overcome that issue, it was decided to open one cell out of the ten to make sure the pressure inside the model is as close as possible to ram-air pressure. The inlet thus created has the following configuration: closed at the rib and a maximum of 7.5% of chord height reached at mid-cell. The rest of the cells are closed and remain sealed. This configuration is likely to generate slightly more drag than it would with all cells sealed, as there is one inlet on the leading edge. Basically, 90% of the model is supposed to behave like it would in an inflated sealed configuration, with a difference for the remaining 10% because of the inlet. For comparison, the performance (lift, drag and pitching moment) was measured with all cells opened as well.



Figure 22: One-intake configuration: (a) front; (b) trailing edge

As seen in figure 22, opening one cell results in overall better inflation. The ripples that appeared from $\pm 3^\circ$ in the previous configuration are absent from this configuration. However, as depicted in figure 22a, the frontal collapse at the leading edge is still present because the incoming air flow in the stagnation point region has slightly higher pressure than the inside air, resulting in frontal collapse and loss of performance. Because of the stiffness of the taped leading edge, this phenomenon is even worsened and affects a region which is approximately 5% of the chord length, resulting in a triangle equilibrium shape, already observed in figure 20a.

The graphs presented in figure 23 show the results obtained for the one-intake configuration, as well as the all-opened cells configuration, where all cells are opened with the height at middle-cell being 7.5% of chord (see figure 24 for photograph). They are also compared to the results obtained for the side-inflated configuration. The lift performance is almost identical for the three configurations. This is to be expected as the surfaces generating lift in all configurations are the same. Indeed, the leading edge in the one-intake and side-inflated configurations undergoes frontal collapse, resulting in a dramatic loss of lift in that region which produces almost no lift. The major difference is the stall behaviour. Stall is reached at 10° for the all-open intakes configuration while the performance is much better with only one inlet open, where stall is reached at 14.2° . This can be explained by

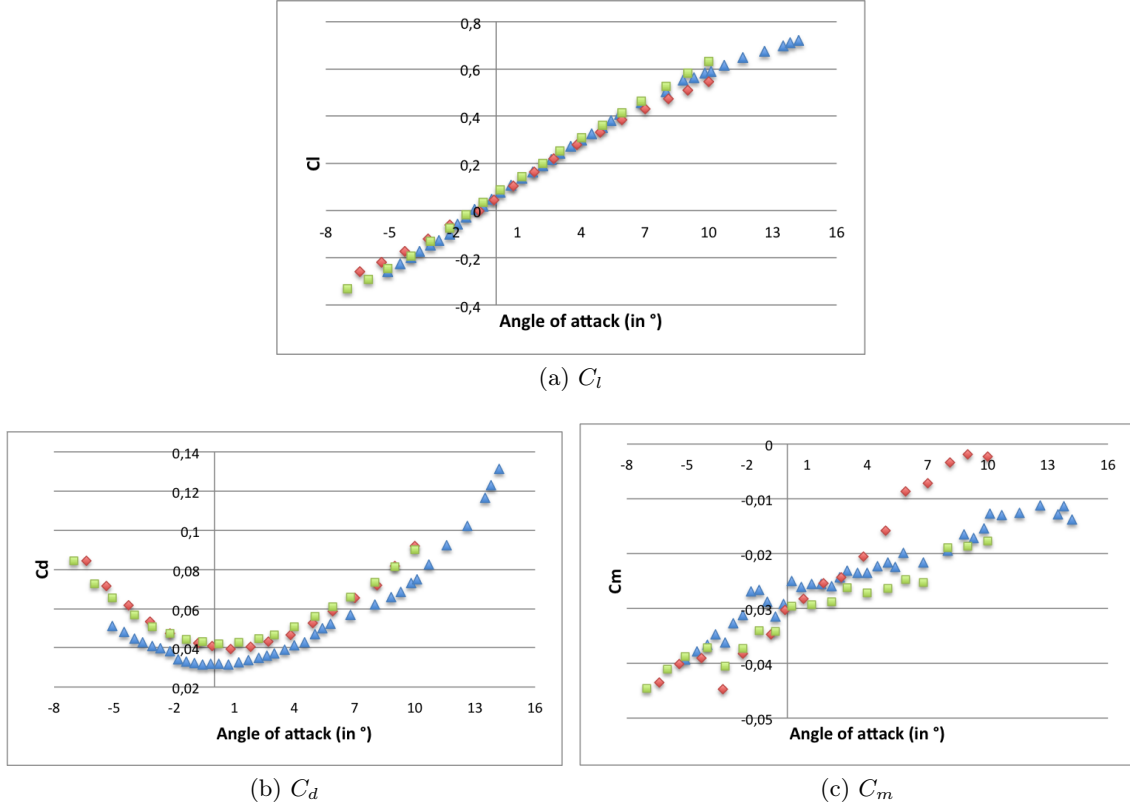


Figure 23: Performance of one-intake configuration (blue triangles); all-open cells (red diamonds); and side-inflation configuration (green squares)

two phenomena: first of all, the sharp lips of the intakes cause flow detachment at lower angles of attack than for the sealed leading edge configuration, resulting in stall at lower angles of attack. Secondly, these openings are not calculated for operating at high angles of attack. Therefore, at high angles of attack, the intakes lips start collapsing, ending up perturbing the flow in the intake and causing instability.

As for the drag force, it is much higher in the all-open cells configuration, because of the presence of nine additional intakes, which are great sources of drag as discussed in section 2.2.3. The 7° angle of attack is to be used for comparison between the different configurations as it is a common flying angle of attack for ram-air parachutes. At that angle, there is a 16.2% increase in drag between the one-intake configuration and the all-open cells configuration, due to the presence of nine additional air intakes in the latter. In comparison, the decrease in lift is only 1.7% at that angle of attack between these configurations. The one-intake configuration has much better performance in terms of drag compared to the side-inflated configuration as well.

Note that the pitching moment curves in figure 23c show a change in slope at 9.5° for the all-open cells configuration and at 14.5° for the one-intake configuration, matching the lift coefficient change in slope and the stall behaviour experimentally observed to be respectively at about 10° and 14.3° angles of attack for both configurations.

The one-intake configuration is to be used as basis for comparison for the rest of the results: 90% of the cells seems to behave as they would in a properly inflated sealed leading edge configuration with only 10% behaving differently. It is very interesting to note the good

performance of this configuration compared to the all-open cells configuration. There is no noticeable difference in inflation between the two, whereas only 10% of the leading edge is open in the former compared to it being open along the whole span in the latter. This suggests that correct inflation can be obtained with only a small portion of the leading edge being opened, with significant improvement on the aerodynamic performance, as a consequence of the decrease in drag.

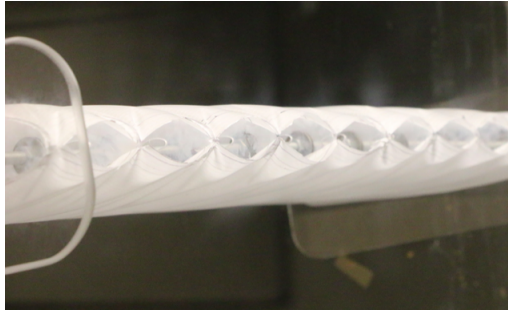


Figure 24: Open-cells configuration

6.3 Influence of trailing edge deformation

Unlike rigid wings, the shape of the trailing edge differs greatly from the aerofoil shape for ram-air wings, as internal pressure causes cloth expansion at the trailing edge. This has been identified in previous literature as a potential important drag mechanism as the flow properties at the trailing edge are greatly changed. In order to quantify the significance of this drag mechanism in the total drag generated by a ram-air wing, the trailing edge was constrained on a length equal to 15% of the chord on both upper and lower surfaces. The leading edge configuration used for this set of experiment was the all-opened cells configuration.

At first, the trailing edge was taped on both upper and lower surfaces on a length corresponding to 15% of the chord, as seen in figure 25b. The idea is to limit the deformation of the trailing edge during the inflation process. However, the trailing edge is not much constrained, even though that configuration presents the advantage of ensuring better sealing of the ram-air wing and therefore better pressurization. In order to make sure that the trailing edge was constrained, a second configuration was studied with stitching threads constraining the upper and lower surfaces. Figure 25c shows the comparison between stitched cells and a free cell (the middle cell in the picture) where the thread broke and the trailing edge is free. The difference is noticeable as the ballooning of the trailing edge is clearly diminished with the threads.

The results for these configurations are presented in figure 26. These results suggest that the trailing edge ballooning effect only accounts for a small amount of the total drag. There is no major improvement in drag performance. The 7° angle of incidence was used for comparison, as it is the angle for which the best lift to drag ratio is obtained for all

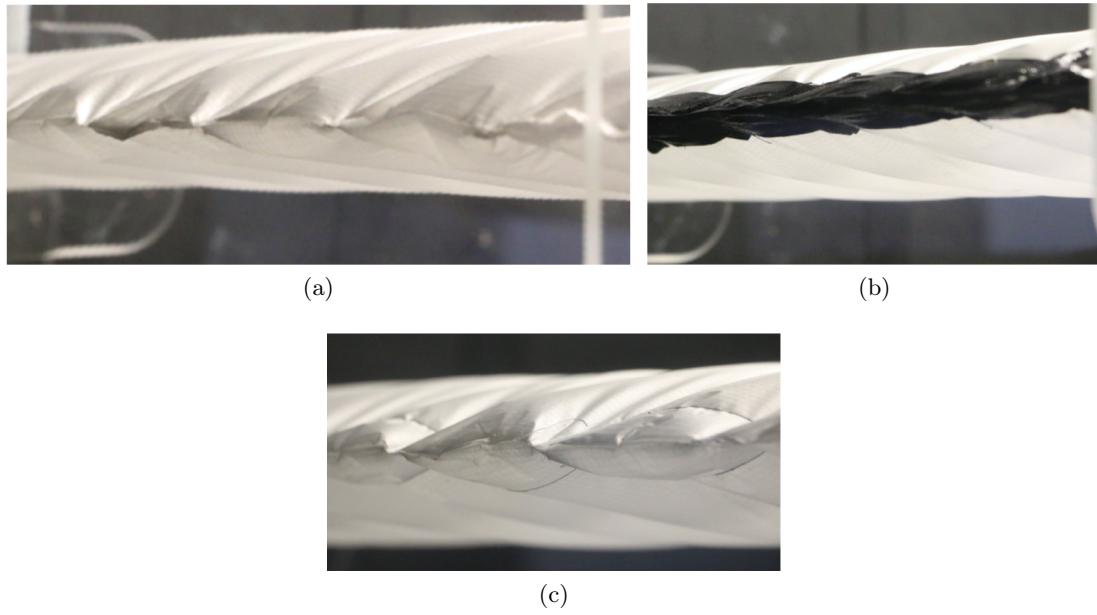


Figure 25: All-opened cells configuration trailing edge: (a) free; (b) with tape; (c) stitched with threads and middle free cell

configurations and a common operating angle for paragliders. At this angle, there is only a 1.3% decrease in drag when the trailing edge is stitched compared to when it is left free, and a 2.4% decrease in drag when it is taped compared to the original state. Note that the stitched trailing edge configuration does not generate less drag than the taped configuration, suggesting that leakage at the trailing edge plays an identical role as trailing edge deformation in drag.

In terms of lift, the performance of the stitched configuration presents slightly better lift properties than both the taped and free leading edges configurations. The gain in lift is about 4.6% at 7° between the stitched and the free trailing edge configurations. This might be explained by the fact that, with the threads, the trailing edge maintains a shape closer to the aerofoil shape and thus generates a bigger amount of lift.

Overall, the trailing edge ballooning effect due to inflation, only slightly decreases the aerodynamic performance. Stitching the trailing edge only provides a 5.6% increase in maximum lift-to-drag ratio compared to the original free trailing edge configuration. This experiment has some limitations as the trailing edge was not properly constrained. However, properly constraining the trailing edge might constitute a technical challenge for ram-air parachutes manufacturers and might not be the best way to optimise aerodynamic properties as the decrease in drag is relatively small.

6.4 Full-range inlet configuration

This section aims at identifying the significance of the leading edge configuration as a drag mechanism. The results presented in this section were obtained with the "full-range" leading edge cut, designed to cover all angles of attack in the -4° to 15° range, as was

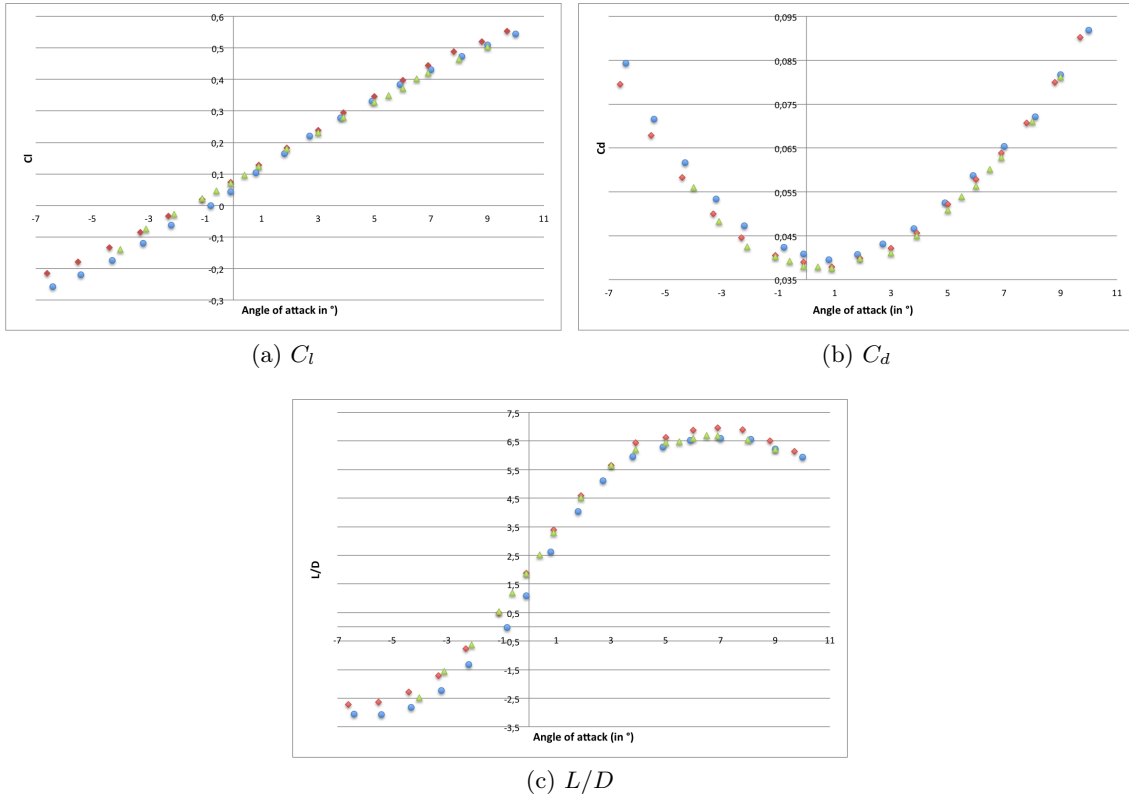


Figure 26: Performance of all-opened intakes configuration with free leading edge (blue circles), taped leading edge (green triangles) and stitched trailing edge (red diamonds)

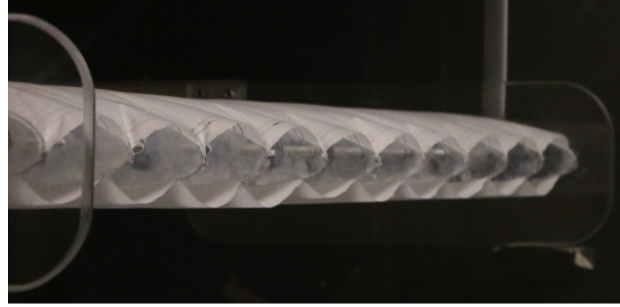
described in section 3.3.

6.4.1 Initial configuration and improvement

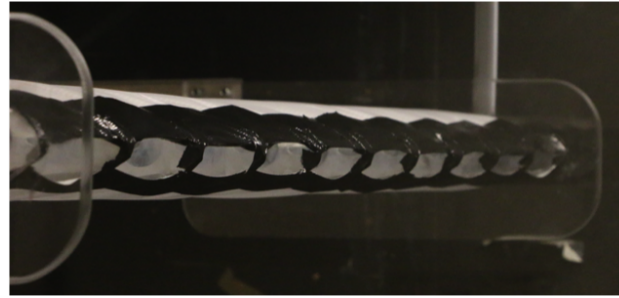
The inlets were cut in all cells according to the calculations done in XFOIL. The cells, at rest, are cut like shown in figure 9a. Figure 27a shows the shape assumed by the inlets once the model is inflated. This shape is different than what was calculated. This is explained by the following: firstly, because the rib spacing is fixed in the model, an extra length of cloth had to be added in order to reproduce the scalloping effect of real ram-air wings along the chord: this results in a catenary shape of the leading edge lips which is amplified compared to real ram-air wings (see figure 28). This is especially true regarding the upper lip. Secondly, in ram-air parachutes, suspension and internal lines tend to increase the stiffness of the inlets. The inlet obtained is therefore bigger than designed, resulting in a poorer performance than expected, as shown in figure 29. In order to reproduce the proper behaviour of the inlet, tape was added at the leading edge, therefore stiffening it and ensuring that the inflated shape of the inlet matches that of real ram-air parachutes. The effect of taping the leading edge is seen in figure 27b, where the inlet shape is closer to what was designed. There is a small scalloping in-between ribs which is a consequence of the inflated shape and matches full-scale ram-air parachutes inlet behaviour.

Both configurations are oriented at 48° with regards to the chord. However, the initial configuration leading edge opening can reach at its maximum 44.5 mm in height, which

corresponds to 14.8% of the chord, whereas the improved (taped) configuration leading edge maximum height is 13 mm which equals 4.3% of the chord. The height was designed to be 3.5% of chord. Due to inflation, the upper and bottom lip both assume the shape of a catenary (though this phenomenon is much reduced in the improved configuration), which explains the difference in height at half-cell. That difference is only 0.8% of the chord for the improved configuration.



(a) Free leading edge



(b) Taped leading edge

Figure 27: Full-range configuration



Figure 28: Ram-air parachute inflated leading edge. Adapted from Mpora website
<https://mpora.com/windsurfing-kitesurfing/kitesurfing-equipment-beginners-guide-essential-gear#BuLdV9cwJuTwllGs.97>

6.4.2 Results of the full-range configuration

The results of the first "full-range" inlet configuration are presented in figure 29. They are compared to the results of the base configuration (one-intake configuration). The difference between the two configurations (free leading edge and taped leading edge) is not significant in terms of lift, except at high angles of attack (cf. figure 29a). Restricting the inlet size

with tape has resulted in a higher maximum lift coefficient. It reaches 0.61 while it stays below 0.52 for the free leading edge configuration. The better performance at high angles of attack is easily explained: the peak of low pressure at the upper lip is absent with the free leading edge configuration because of the rounded shape on the inlet. This rounded leading edge shape also explains why C_l stays positive even for negative angles of attack for the free leading edge configuration. The lift performance of both these configurations is poorer than for the base configuration. This gap in performance increases as the angle of attack is raised which is explained by the loss of lift from the separation bubble on the bottom surface caused by the redirection of flow downwards out of the inlet.

Stall is reached at approximately 13.6° for the free leading edge configuration and at 14° for the taped leading edge. The change in slope in the pitching moment for the free leading edge configuration occurs at 10° . It is matched by a kink in the drag coefficient slope. The same changes in slope occur for the taped leading edge configuration at 12° . Stall occurs slightly afterwards.

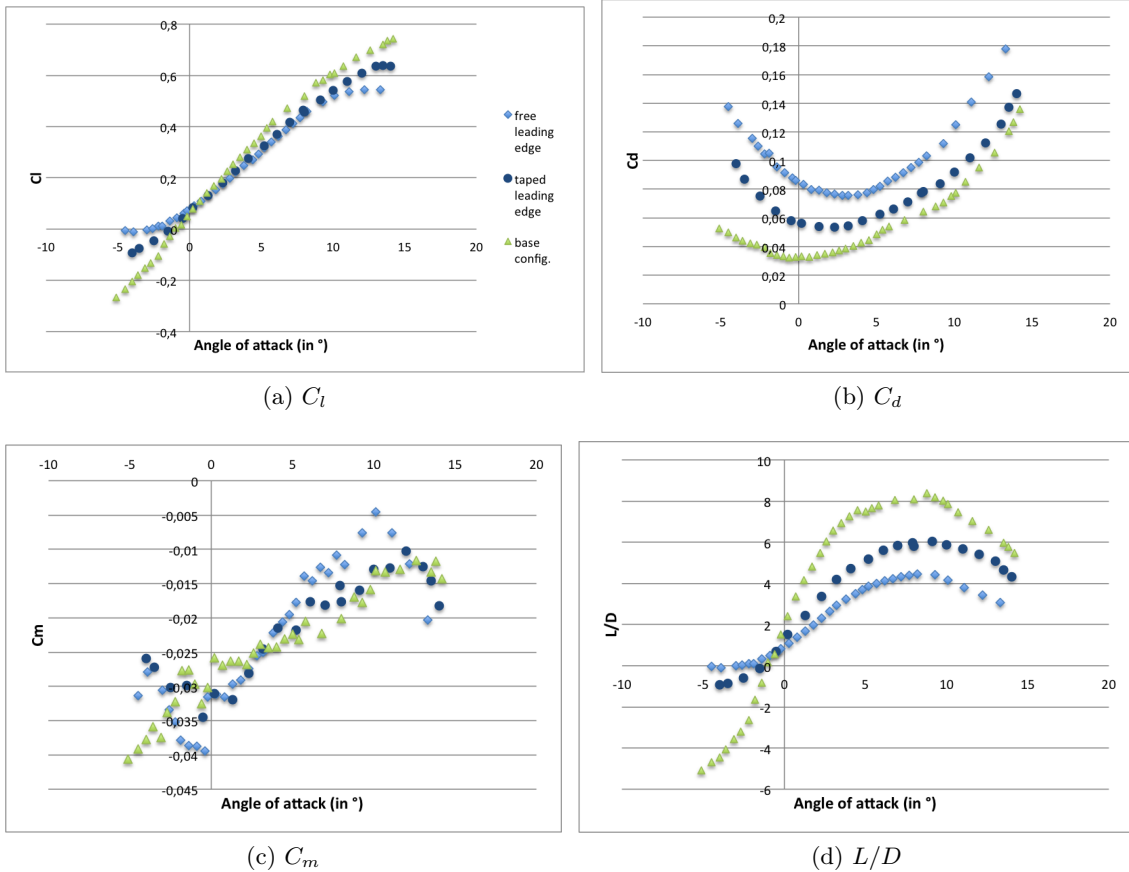


Figure 29: Full-range intake configuration results

With regards to drag, there is a very significant difference between the three configurations, providing evidence that the leading edge inlet indeed plays a predominant role in generating drag. There is a 20% increase in drag between the base and the taped leading edge full-range configuration at 8° and an additional 30% increase between the taped leading edge

and the free leading edge full-range configurations at that angle. This is due to major flow detachment caused by the presence of the inlet on the leading edge. Flow separation is further increased by the size of the inlet. Note the shift in minimum drag as the inlet size is increased: it is reached at approximately 0° for the base configuration, and 2.5° and 4° respectively for the taped leading edge and free leading edge "full-range" intake configurations. This is due to the size of the inlet: one possible explanation is that the angle at which the least amount of air is able to penetrate inside the parachute canopy depends on the size of the inlet. The dramatic deterioration in aerodynamic performance in terms of drag is reflected by the lift to drag ratio curve. The maximum ratio is above 8 for the base configuration while respectively 6 and 4.3 for the taped and free leading edge "full-range" intake configurations. The maximum lift-to-drag ratio is obtained at around 8° angle, which matches the design of the aerofoil, conceived to provide best performance at relatively high angles of attack, which ram-air parachutes usually fly at.

6.5 Optimised inlet and improved configuration

The inlet was cut in all cells according to the calculations done in XFOIL, attempting to give the leading edge the shape presented in figure 9b. However, the same problem arose as in section 6.4: because the model was manufactured as a semi-rigid model and an extra length of cloth had to be added in between ribs, the inlet is much less stiff than real ram-air parachutes inlets, resulting in an accentuated scalloping of the inlet.

Thus, like it was the case in section 6.4, two configurations were studied for the optimised inlet configuration: a free leading edge version, shown in figure 30a and a taped leading edge version, shown in figure 30a, which is closer to what was designed in section 3.3. Both configurations are oriented at 42° with regards to the chord. The free leading edge configuration gives a maximum inlet height at half-cell of 34.8 mm (11.6% of the chord), while the maximum inlet height for the improved configuration (with tape) is 9.4 mm (3.1% of the chord). It was designed to be 9.2 mm (3% of the chord).

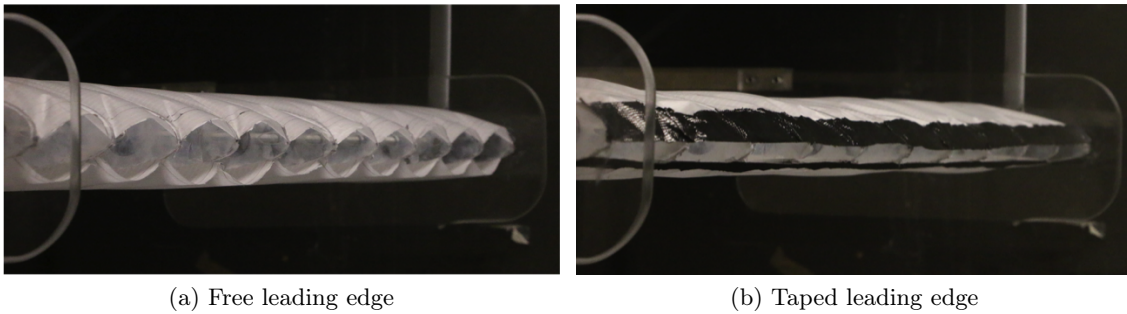


Figure 30: Optimised intake configurations

Figure 31 sums up the results obtained for all inlet configurations, compared to the base configuration with one open intake. The inlet sizes and orientations with regards to the chord are detailed previously in sections 6.4.1 and 6.5 and summed up in table 4.

The first interesting result to note is the good performance of the optimised inlet at high

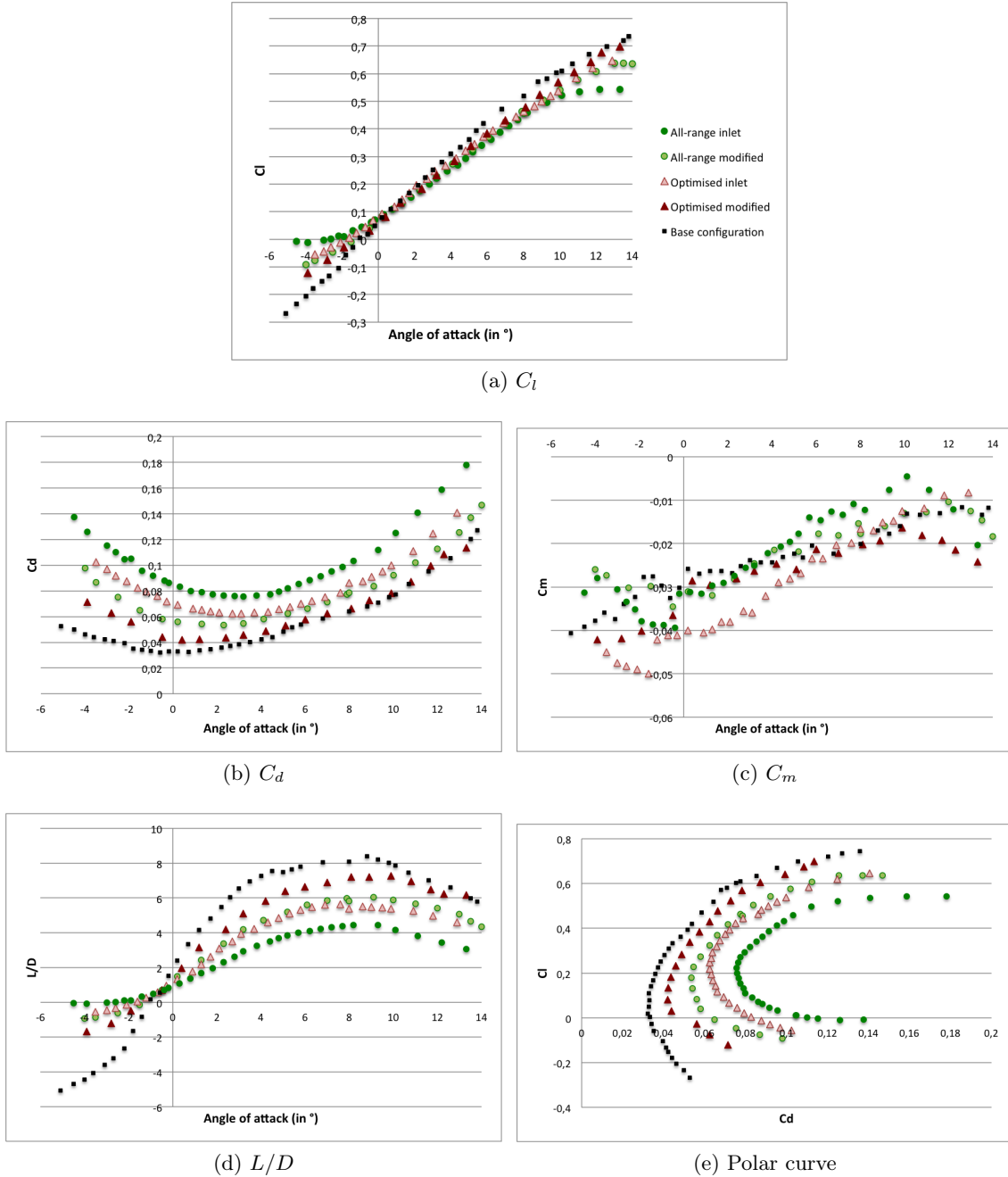


Figure 31: Performance of all inlet configurations compared to base configuration (black squares)

angles of attack. According to its design on XFOIL, for this inlet configuration, the stagnation point is located outside of the inlet (on the airfoil section) when the angle is out of the $[0^\circ; 11^\circ]$ range. This might have caused instability at the leading edge. However, the optimised inlet configuration stalls at 13° , while the modified (taped) optimised inlet configuration was found to stall at 13.3° which is higher than expected. Likewise, these optimised configurations were able to reach about -4° before becoming unstable. This shows that using the optimised inlet configuration is indeed possible for kite manufacturers, as it provides good lifting properties on the commonly-used range of angles of attack. The lift performance for all configurations is almost identical from -2° to 10° , except for

the base configuration which is overall slightly better. For higher angles of attack, the optimised configurations, with free and taped leading edge, have better performance but stall earlier. This can be explained by the fact that the recirculation bubble formed on the bottom surface increases with inlet size, as more fluid is pushed downwards, resulting in loss of lift from the bottom surface. However, the "full-range" inlets were designed to cover a greater range of angles of attack, and they stall at higher angles than the optimised configurations.

Figure 31b shows a clear correlation between inlet size and generated drag. As seen in table 4, the amount of drag increases with inlet size. It is interesting to note that the modified (taped) optimised configuration has very similar drag performance compared to the base configuration, especially at higher angles of attack. This suggests that the flow behaviour for a very small inlet is the same as for a collapsed leading edge: it is likely that the amount of air able to actually penetrate into the inlet is very small, and is being instantly pushed downwards, towards the leading edge inlet bottom lip.

Figure 31d shows the lift-to-drag ratio for all configurations. This figure shows a similar correlation between inlet size and decrease in lift-to-drag ratio. As shown in table 4, at 8° angle of attack, the modified optimised configuration lift-to-drag ratio is 89% of the base configuration. For the biggest inlet, it is only 55%, meaning that the performance has been divided by a factor of 2.

Configuration	Inlet height	Orientation	C_l		C_d		L/D	
			Value	Of base	Value	Of base	Value	Of base
Full-range: free LE	14.8%	48°	0.453	87%	0.102	159%	4.43	55%
Full-range: taped	4.3%	48°	0.457	88%	0.0785	122%	5.82	72%
Optimised: free LE	11.6%	42°	0.464	89%	0.0861	134%	5.39	67%
Optimised: taped	3.1%	42°	0.473	91%	0.0659	103%	7.17	89%

Table 2: Results for all inlet configurations at 8° angle of incidence

7 Discussion

7.1 Sources of imprecisions

This section aims at discussing the potential limitations of this experimental study. The principal sources of errors were accounted for to check that the results were obtained with enough reliability.

Wind tunnel measurement imprecisions

The wind tunnel equipment precision was given in section 5.2.2. The speed measurement precision can amount to 0.2% error in dynamic pressure and therefore in all aerodynamic coefficients. The force balance precision depends on the forces measured: it can result in 0.6% error in drag, 2% error in lift and 3% error in pitching moment.

Furthermore, it is likely that a small amount of lift was transferred to the back strut and

therefore not measured by the force balance, which measures lift and drag with the front mountings, and pitching moment with the back mountings. However, the front struts were positioned at 25% of chord where most of the lift is generated, while the back strut was positioned about 750 mm (250% of chord) behind the front struts. Under the assumptions that about 70% of the lift is generated by the front 50% of the wing, and that the trailing edge produces a negligible amount of lift, which is consistent with figure 6 in section 2.2.2, and using a simple beam model, then at least 98% of the lift should be captured by the front mountings. This adds less than 2% in uncertainty in lift.

Finally, as discussed in section 5.2.5, the blockage ratio of the whole setting (struts, rods and end-plates included) is 4.9%. As it is below 5%, no correction factor was applied to the results obtained in section 6, based on conclusions by Choi and Kwon [17]. However, this might result in very small errors for the aerodynamic coefficients.

Overall, all wind tunnel setting imprecisions lead to a maximum of 4.2% error in lift, 0.8% error in drag and 3.2% error in pitching moment.

Manufacturing imprecisions

The previous paragraph listed the errors inherently linked to the wind tunnel setting and the force captors precision. This paragraph focuses on the errors originating from manufacturing the semi-rigid model, and particularly in gluing the cloth on the rigid ribs. Hand-manufacturing the semi-rigid model resulted in a precision of ± 1 mm (0.3% of chord) in inlet cuts and cloth gluing. These human errors particularly impacted the leading edge. First of all, for the base configuration, the leading edge could not properly be sealed with one layer of cloth as was intended. Taped had to added, stiffening the leading edge and might have had an impact of the leading edge equilibrium shape and thus on the flow. In addition, not all inlets were exactly the same size, as a result of hand-manufacturing imprecisions. Overall, this might have been balanced by the fact that some inlets were slightly larger than intended, while some were smaller.

These hand-manufacturing imprecisions do not affect measurement imprecisions and were kept to the minimum. Furthermore, it should correctly reflect real-scale parachutes manufacturing errors.

7.2 Overall performance compared to previous results

The previous paragraph explored the main sources of errors due to experimental manipulation in this project. However, the errors were kept at an acceptable percentage. This section aims at comparing the results obtained in this project to results available in literature. The modified "full-range" inlet configuration is the fittest to be compared with past literature as the inlet size, about 7.5% of chord, is similar to standard inlet size used in previous works, between 8% and 10% of chord. The Reynolds number used in these studies is $Re = 10^6 - 2 \times 10^6$.

Figure 32 shows results obtained for a semi-rigid ram-air wing with standard inlet size by Nyren et al. [5], compared to results found in this project for the modified (taped leading edge) "full-range" inlet configuration. The lift, drag and lift-to-drag ratio curves behave

similarly. However, the lift performance of the semi-rigid model used by Nyren et al. is better, with a 20% difference at 7.5° . One possible explanation is that Nyren et al. used a 3.5 aspect ratio model, which generated more lift. Overall, as seen in table ??, the lift performance produced by this project model is poorer than previous experimental projects. Ware and Hassel [8] found a lift coefficient of 0.58 for their scaled flexible model (aspect ratio of 3 and 10% of chord inlet size) compared to 0.45 for this project. This can be explained by canopy deformation mechanisms due to line tension which produce more lift but also additional drag (for instance, trailing edge deflection and chord-wise reduction when increasing the angle of attack).

On the other hand, the drag coefficient measured for the modified "full-range" inlet con-

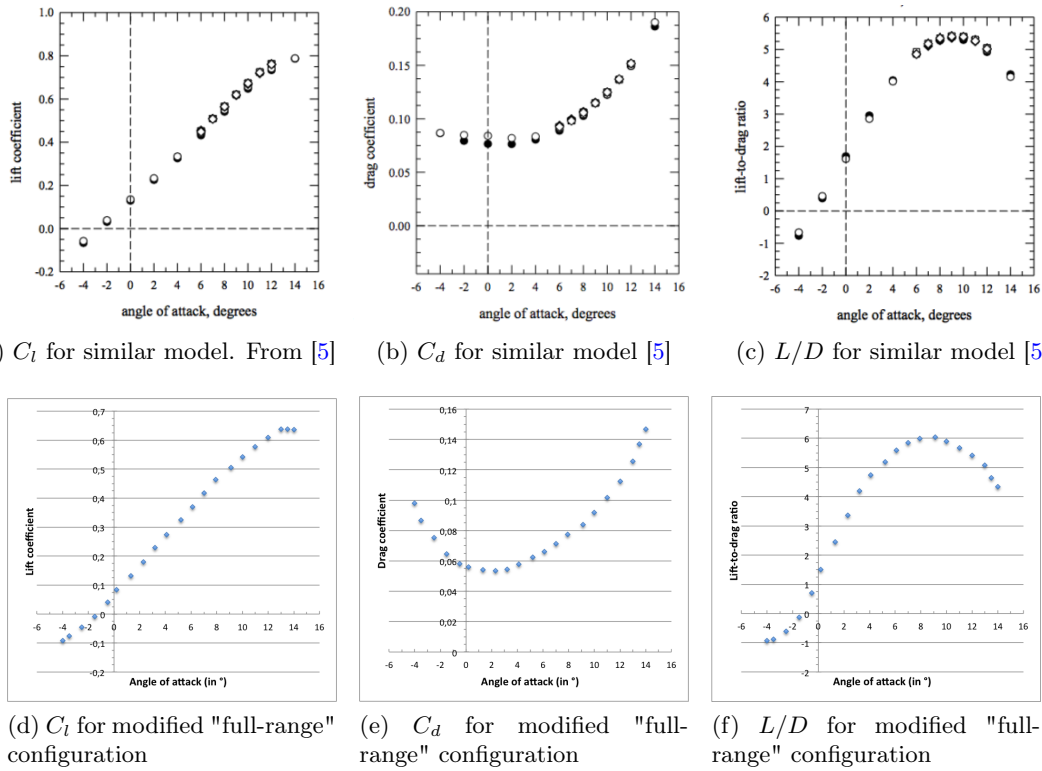


Figure 32: Comparison between modified "full-range" configuration results and values obtained on a similar semi-rigid model [5]

figuration is considerably lower. This is explained by the smaller size of the inlet (7.5% of chord compared to approximately 10% for the configurations used for comparison), which greatly reduces drag, as well as the absence of three-dimensional deformation mechanisms which occur for Ware and Hassel's flexible model [8] well as Fogell's numerical simulation [3] and increase flow obstruction.

7.3 Impact of trailing edge deformation as a drag mechanism

Due to inflation, the trailing edge puffs out along the span (except at the ribs). This phenomenon previously described in literature, was suspected to have a significant impact on aerodynamic performance. Fogell suspected the rounding up of the trailing edge could

Configuration	Angle of attack	C_l	C_d	L/D
Semi-rigid [5]	7.5°	0.53	0.1	5.25
Flexible [8]	7.5°	0.58	0.135	4.15
Project modified "full-range"	7.5°	0.45	0.075	6
Numerical [3]	5°	0.306	0.132	2.32
Project modified "full-range"	5°	0.3	0.06	4.8

Table 3: Project modified "full-range" configuration results compared to previous literature results

be an important drag mechanism [3]. Previous numerical works on 2D models have found this phenomenon to have a considerable impact on lift and drag: Ross [6] found that it had a significant effect on the lifting capacity of the aerofoil. Jones and Ames found that the drag coefficient was increase by 0.02 when rounding up the trailing edge on a NACA 4412 airfoil [18].

However, the results found in this project indicate that this phenomenon only has a minor influence on the aerodynamic performance of a full ram-air 2D wing. The maximum lift-to-drag ratio (obtained for approximately 8°) was increased by 5.6% by stitching the trailing edge on a region long of 15% of the chord. The impact on lift was small, with a 2% increase in performance when stitching the trailing edge. The increase in drag coefficient was actually about 0.002 compared to 0.02 found by Jones and Ames [18]. The difference between this project and previous numerical works on 2D airfoils is that the flow around ram-air parachutes is clearly different from the flow around airfoils, due to the introduction of the inlet cut. Around the trailing edge upper surface, the flow is detached. Therefore, the trailing edge shape is not as significant on lift and drag performance as for the airfoil. These results indicate that the trailing edge should not be the main point of concern for kite manufacturers in developing better performing kites: first of all because the impact is minor, and also because of the technical challenge of trying to properly constrain the leading edge ballooning without affecting the flow otherwise.

7.4 Impact of inlet size on drag

Trailing edge ballooning was found to be a minor drag generating mechanism in this project. On the contrary, the experimental study of different inlet sizes confirmed what recent works in ram-air parachutes literature suggest: leading edge intakes are major contributors to drag. This project results show that changes in inlet size have considerable impact on the aerodynamic performance and particularly on drag, while the lift performance was not significantly affected.

Figure 33 shows the results obtained in this project for the base, the modified optimised inlet and the modified "full-range" inlet configurations compared to results produced by Ross [6] for experimental 3 aspect ratio kites at $Re = 10^6$. The lift-to-drag ratios obtained for this project were overall greater compared to Ross's results, but this is easily explained as line drag was taken into account in his results. Ross found that diminishing the inlet size helps with the lift performance as the lift coefficient can go higher. The difference was very clear between the 4% inlet and the 8% inlet, going from a maximum lift coefficient

of 0.75 to a 0.55. In this project, the same trend was noticed as in Ross's work [6] and Belloc et al.'s work [12], though the discrepancy between the different inlet configurations was less pronounced. Ross found that a 50% decrease in inlet size (from 8 to 4% of chord inlet size) results in a 25% increase in maximum lift-to-drag ratio. In this project, it was found that a 33% decrease in size results in a 20% increase in maximum lift-to-drag ratio, mainly caused by the decrease in drag coefficient. As suggested earlier, this behaviour is partially accounted for by the fact that when the inlet size is reduced, a smaller amount of air is able to penetrate the intake, therefore a smaller amount of air is diverted downwards out of the inlet, resulting in a smaller recirculation bubble on the bottom surface.

This project confirmed that working on leading edge optimisation is a very promising way in improving aerodynamic performance. Smaller inlets might provide sufficient inflation while greatly reducing drag and performing at higher angles of attack.

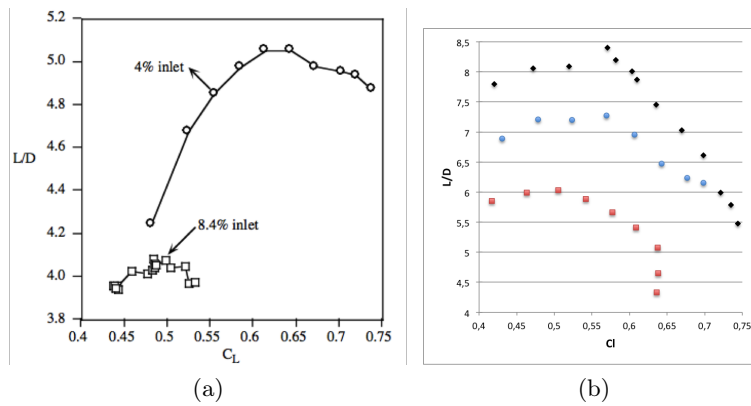


Figure 33: Comparison of lift-to-drag ratio obtained for different inlet sizes: (a) numerical results from Ross [6]; (b) experimental results from this project (black: one-intake configuration; blue: modified optimised configuration 5% inlet size; red: modified "full-range" configuration 7.5% inlet size)

7.5 One-intake configuration

A very interesting point of this project is the aerodynamic performance of the one-intake configuration, where one cell out of the ten was opened to ensure inflation while the others stayed sealed. The results obtained suggest that such a configuration gives better aerodynamic performance than all the other configurations studied in this project and that such a leading edge configuration should be looked into to further optimise aerodynamic performance of ram-air parachutes. No noticeable difference in inflation was observed between this configuration and the others, suggesting that opening 10% of the leading edge might ensure sufficient inflation. This configuration best lift-to-drag ratio was 8.4 compared to 7.3 for the second best configuration (modified optimised inlet), which means a 15% increase. The lift performance was better than all configurations and the drag performance better than all except the modified optimised inlet which produces similar drag force.

Very few works in available literature discuss such leading edge configurations. Bergeron et al. looked at the difference in lift and drag coefficients for rigid canopies with a round nose

(original CLARK-Y airfoil) and flat nose with the same orientation as standard inlets but no opening in the leading edge [11]. The models in these works were rigid, therefore looking at aerodynamic characteristics without considering the inflation issue. The interesting results they found was that the flat nose configurations had very similar lift and drag curves compared to the original round nose configurations with no significant difference in drag nor lift until 10° . After that angle, the lift performance of the flat nose configurations were slightly poorer than the original configurations, although stall was reached at the same angle [11]. This can help understand why the one-intake configuration performed better than all other configurations tested in this project. The leading edge was flattened for the one-intake configuration for all sealed cells because pressurization inside the cell was not enough to maintain a round nose against the incoming flow of air. Thus the leading edge could be likened to the flat nose configurations in Bergeron et al. study. This configuration gives similar lift to the aerofoil-shaped configuration and only a slight increase in drag. The increase in drag is due to flow separation at the top and bottom of the leading edge due to sharper angles than with a round aerofoil. However, the air is not able to penetrate inside the ram-air, and therefore there is no significant separation bubble due to recirculation on the bottom surface. This results in less drag and explains why this configuration performs well despite the flattened shape of the closed cells.

8 Conclusions and recommendations

The objectives of this project were fulfilled. A semi-rigid model was built which permitted to isolate the main drag mechanisms and study specifically the influence of the inlet size, as well as the influence of trailing edge deformation due to inflation on aerodynamic performance and more precisely on drag. Examples of such semi-rigid models, with rigid internal structure - ribs - and flexible cloth, are very rare in ram-air parachutes literature. A rigid model would have been easier to manufacture and to compare to previous works. However, rigid models do not properly capture the mechanisms occurring at the air intake because of inflation and flow perturbation. Therefore a semi-rigid model was used, presenting the advantages of decoupling drag mechanisms and isolating the influence of the inlet. This semi-rigid model was designed to properly reproduce real-scale ram-air parachute air intake behaviour and air-flow, despite its reduced scale: 0.3 m chord and 0.755 m span. To emulate the air-flow behaviour around ram-air parachutes, the Reynolds number used was $Re = 5 \times 10^5$. The model was tested in a 1.22 m by 1.37 m wind tunnel section and the lift, drag and pitching moment coefficients were measured for the different inlet and trailing edge configurations.

The results obtained from these wind tunnel experiments were overall comparable to previous literature in terms of lift and drag performance. The better drag performance is explained by the absence of canopy lines, as well as the fact that other drag mechanisms were suppressed, such as span-wise scalloping of the canopy, trailing edge deflection, chord-

wise deformation. The results obtained seem to indicate that preventing the trailing edge from rounding-up in between ribs due to inflation yields minor increase in aerodynamic performance, of about 5%. Coupled with the technical difficulties in restraining the trailing edge without triggering other effects, the limited change in aerodynamic performance suggests that working on trailing edge geometry is not the most efficient way to optimise paragliders aerodynamics.

On the other hand, results obtained in this project for different inlet configurations confirm previous studies conclusions that the leading edge air-intake configuration is the most significant contributor to drag for ram-air parachutes, even though the lift performance is only slightly affected. This project showed that variation in inlet size by 2.5% results in drag increase of about 25%. Thus focusing on the leading edge air-intakes configuration is indeed a promising way to optimise ram-air parachutes aerodynamic performance. The optimised inlet configuration which was tested in this project, whose size is reduced compared to standard industrial inlet sizes, indeed proved to provide similar lifting properties with a great decrease in drag.

Very interestingly, this project showed that another type of leading edge configuration could work. The performance of the model was measured when the air intake occupied only 10% of the leading edge. Such a configuration was able to perform over a similar range of angles of attack as the others, with good inflation properties, and a significant improvement in aerodynamic properties (both in lift and drag). was c interest of one-intake configuration. Such configurations, with only partial opening on the leading edge should be investigated on real-scale ram-air parachutes, to confirm the improvement in aerodynamic performance, and to verify stability and maneuverability as well as inflation properties. Partially opening the leading edge, while the majority of the cells are closed could constitute an alternative to the recent trends of adding flaps to close the leading edge once inflation is complete, with the detriment of adding weight to the total structure.

References

- [1] E. Mearns. High altitude wind power reviewed. <http://euanmearns.com/high-altitude-wind-power-reviewed>. Accessed: 2017-09-04.
- [2] J. Stephen Lingard. Precision aerial delivery system - ram-air parachute design. *13th AIAA Aerodynamic Decelerator Systems Technology Conference*, 1995.
- [3] N. A. T. Fogell. *Fluid-structure interaction simulations of the inflated shape of ram-air parachutes*. PhD thesis, Imperial College London, 11 2014.
- [4] S. F. Hoerner. *Fluid-dynamic drag*. Hoerner Fluid Dynamics, Bakersfield, CA, USA, 1965.
- [5] D. Nyren H. Johari K. J. Desabrais, K. Bergeron. Aerodynamic investigations of a ram-air parachute canopy and an airdrop system. In *23rd AIAA Aerodynamic Decelerator Systems Technology Conference*, 2015.

-
- [6] James C. Ross. Computational Aerodynamics in the Design and Analysis of Ram-Air-Inflated Wings. *NASA Ames Research Center*, 1993.
 - [7] World Energy Council. World energy resources, 2016. *World Energy Council*, 2016.
 - [8] G. M. Ware and J. L. Hassell. Wind-tunnel investigation of ram-air-inflated all-flexible wings of aspect ratios 1.0 to 3.0. Technical report, NASA, 1969.
 - [9] J. D. Nicolaidis. Parafoil wind tunnel tests. Technical report, Air Force Flight Dynamics Laboratory, 1971.
 - [10] F. W. Martin. Parafoil aerodynamic characteristics derived from flight measured suspension system loads. In *15th AIAA/CEAS Aerodynamic Decelerator Systems Technology Conference*, 1999.
 - [11] J. Seidel K. Bergeron and T. McLaughlin. Wind tunnel investigations of rigid ram-air parachute canopy configurations. In *23rd AIAA Aerodynamic Decelerator Systems Technology Conference*, 2015.
 - [12] F. Manara F. Sgarbossa A. Meyer Forsting H. Belloc, V. Chapin. Influence of the air inlet configuration on the performances of a paraglider open airfoil. *International Journal of Aerodynamics*, 5 (n 2)(10):83–104, 2017.
 - [13] E. Puskas. Ram-air parachute design considerations and applications. In *Proceedings of the 8th Aerodynamic Decelerator and Balloon Technology Conference*, pages 255–59. American Institute of Aeronautics and Astronautics, 1984.
 - [14] Teijin fibres mechanical properties. https://www.teijin.com/products/advanced_fibers/aramid/contents/aramid/conex/eng/bussei/conex_bussei_hippari.htm. Accessed: 2017-06-25.
 - [15] R. D. Blevins. *Applied Fluid Dynamics Handbook*. Van Nostrand Reinhold Company, New-York, NY, USA, 1984.
 - [16] E. C. Maskell. A theory of the blockage effects on bluff bodies and stalled wings in a closed wind tunnel. 3400, 1963. ARC.
 - [17] C. K. Choi and D. K. Kwon. Wind tunnel blockage effects on aerodynamic behavior of bluff body. 1:351–364, 09 1998.
 - [18] M. B. Ames R. T. Jones. Wind-tunnel investigation of control-surface characteristics v - the use of a beveled trailing edge to reduce the hinge moment of a control surface. Technical report, National Advisory Committee for Aeronautics, 1942.

A NASA LSA-0417 airfoil coordinates

X	Y	X	Y
0	0	0	0
0,002	0,013	0,002	-0,00974
0,005	0,02035	0,005	-0,01444
0,0125	0,03069	0,0125	-0,02052
0,025	0,04165	0,025	-0,02691
0,0375	0,04974	0,0375	-0,03191
0,05	0,056	0,05	-0,03569
0,075	0,06561	0,075	-0,04209
0,1	0,07309	0,1	-0,047
0,125	0,07909	0,125	-0,05087
0,15	0,08413	0,15	-0,05426
0,175	0,08848	0,175	-0,057
0,2	0,09209	0,2	-0,05926
0,25	0,09778	0,25	-0,06265
0,3	0,10169	0,3	-0,06448
0,35	0,10409	0,35	-0,06517
0,4	0,105	0,4	-0,06483
0,45	0,10456	0,45	-0,06344
0,5	0,10269	0,5	-0,06091
0,55	0,09917	0,55	-0,05683
0,575	0,09674	0,575	-0,05396
0,6	0,09374	0,6	-0,05061
0,625	0,09013	0,625	-0,04678
0,65	0,08604	0,65	-0,04265
0,675	0,08144	0,675	-0,0383
0,7	0,07639	0,7	-0,03383
0,725	0,07096	0,725	-0,0293
0,75	0,06517	0,75	-0,02461
0,775	0,05913	0,775	-0,0203
0,8	0,05291	0,8	-0,01587
0,825	0,04644	0,825	-0,01191
0,85	0,03983	0,85	-0,00852
0,875	0,03313	0,875	-0,00565
0,9	0,02639	0,9	-0,00352
0,925	0,01965	0,925	-0,00248
0,95	0,01287	0,95	-0,00257
0,975	0,00604	0,975	-0,00396
1	-0,00074	1	-0,00783
(a) Top surface		(b) Bottom surface	

Table 4: NASA LS1-0417 airfoil coordinates

B Inlet design - XFOIL study

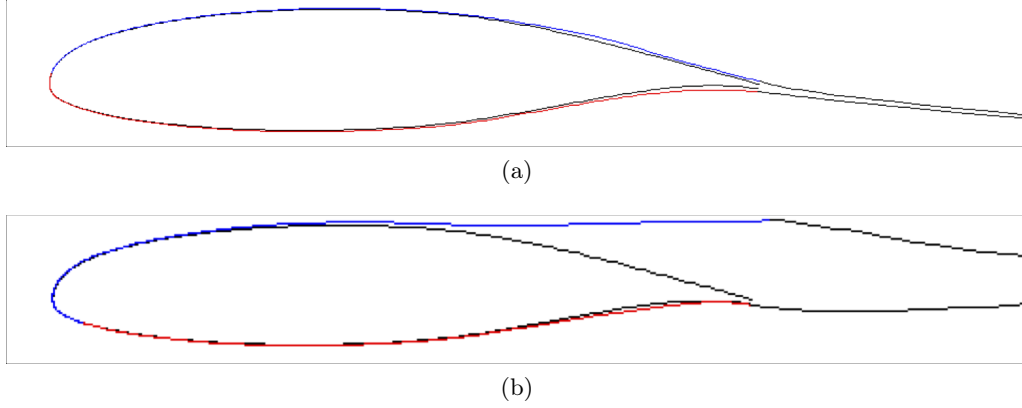


Figure 34: Flow around NASA LS1-0417 airfoil at $Re = 5 \times 10^5$ and at (a) -4° and (b) 15° . Computed with XFOIL

To design the "full-range" inlet, the stagnation point was determined for -4° and 15° angles of attack, using XFOIL. The stagnation point coordinates obtained from the XFOIL analysis are given in table ???. The two first lines give the coordinates used to design the "full-range" inlet, while the two last were used for the "optimised" inlet design.

Alpha	C_l	x (for $C_p = 1$)	y (for $C_p = 1$)
-4	0.0246	0.00266	0.01498
15	1.586	0.04369	-0.03389
0	0.5	0	0
13	1.5	0.03445	-0.03081

Table 5: Coordinates obtained from XFOIL for stagnation points

Note that the difference in stagnation points between the experimental Reynolds number $Re = 5 \times 10^5$ and actual Reynolds number $Re = 1.5 \times 10^6$ is negligible. Figure 35 shows the difference in inlet design between $Re = 5 \times 10^5$, $Re = 7.5 \times 10^5$ and $Re = 1.5 \times 10^6$. When the Reynolds number increases, the stagnation point range very slightly increases as the stagnation point for 15° shifts downwards. But this shift is infinitesimal for such Reynolds number.

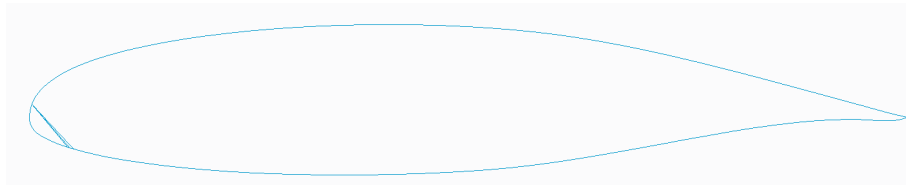


Figure 35: Influence of Reynolds number on stagnation range

C Influence of Reynolds number on flow behaviour around NS(1)-0417 airfoil

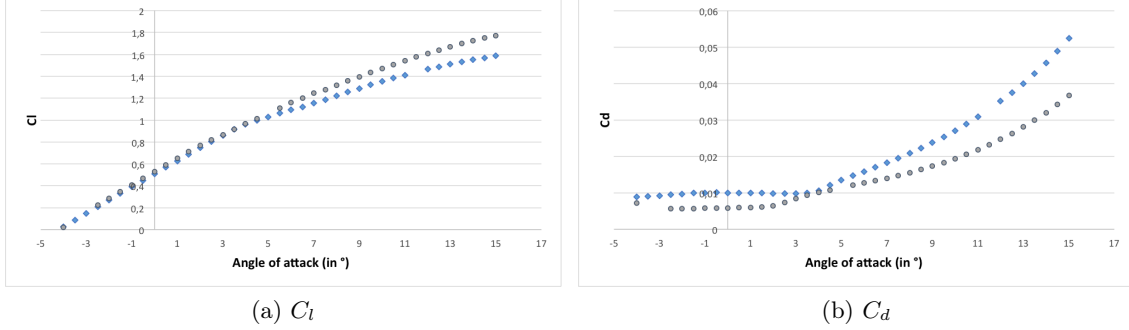


Figure 36: Performance of NASA LS1-0417 at $Re = 5 \times 10^5$ (blue diamonds) and at $Re = 1.5 \times 10^6$ (grey circles)

The air flow behaviour is really similar between $Re = 5 \times 10^5$ and $Re = 1.5 \times 10^6$. The flow stays attached slightly longer in the second case, resulting in slightly better aerodynamic performance: the lift coefficient is very similar in both cases, with slight degradation in performance at $Re = 5 \times 10^5$ after 7° . The drag performance is better at $Re = 1.5 \times 10^6$, as the flows stays attached longer. However, the flow behaves similarly, producing comparable values for lift and drag coefficients. Furthermore, as discussed in section 2.2.2, the flow around a ram-air wing is greatly different to the flow around the aerofoil. Thus, the small differences in drag and lift coefficients for the aerofoil are acceptable, because the flow will be greatly perturbed by the inlet.

D Model drawings

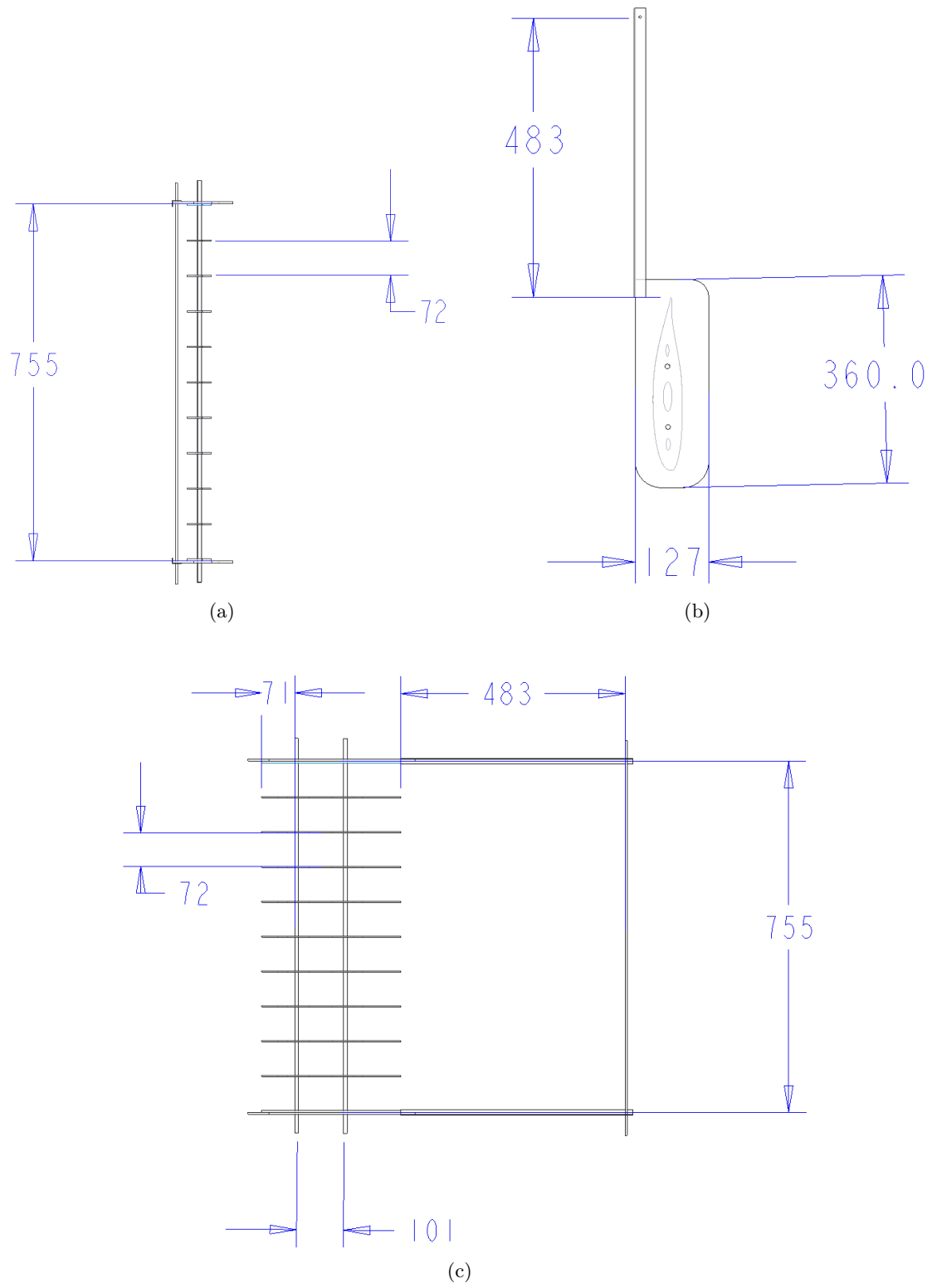


Figure 37: Model drawings seen from: (a) front; (b) right; (c) up

E Determination of cloth fabric Young modulus

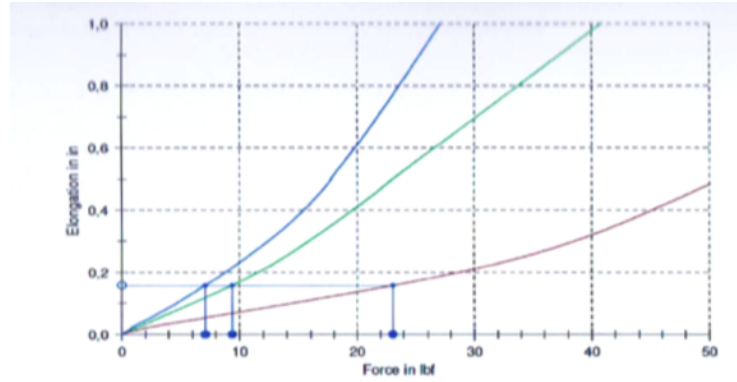
The fabric manufacturer gives the results shown in figure 38 for tensile tests on $50 \times 400 \text{ mm}^2$ samples.

From these figures, the elasticity modulus of the fabric in the 0° direction can be obtained:

$$\sigma_{1\%} = \frac{F_{1\%}}{S} = 0.01 \times E \quad (18)$$

As, $F_{1\%} = 23.1 \text{ lbf} = 102.75 \text{ N}$, the following result is obtained:

$$E = \frac{102.75}{0.01 \times 0.4 \times 0.05} = 514 \text{ MPa} \quad (19)$$



(a) Tensile tests results

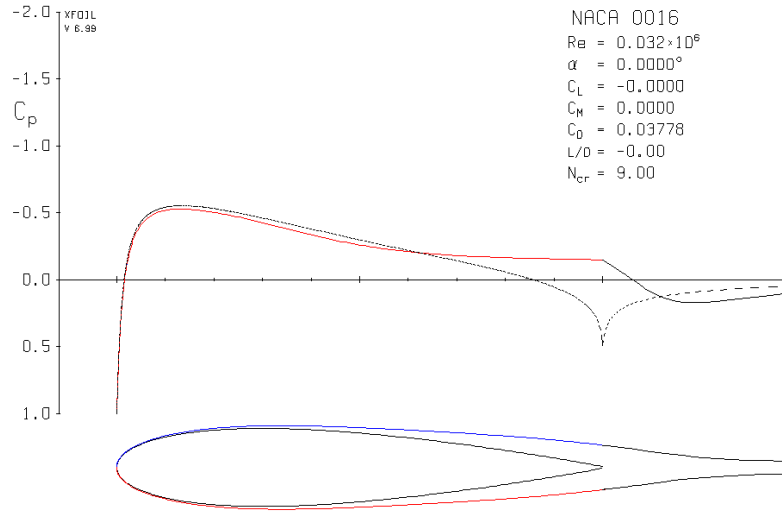
Legende	warp 0°	condition	10 lbf	25 lbf	50 lbf	100 lbf	200 lbf	ε 1% lbf
Nr	1	lab	7,2	17,1	48,3	-	-	23,1
Legende	fill 90°	condition	10 lbf	25 lbf	50 lbf	100 lbf	200 lbf	ε 1% lbf
Nr	2	lab	16,9	55,5	125,8	-	-	9,3
Legende	bias 45°	condition	10 lbf	25 lbf	50 lbf	100 lbf	200 lbf	ε 1% lbf
Nr	3	lab	23,2	88,0	245,7	-	-	7,1

(b) Chart legend

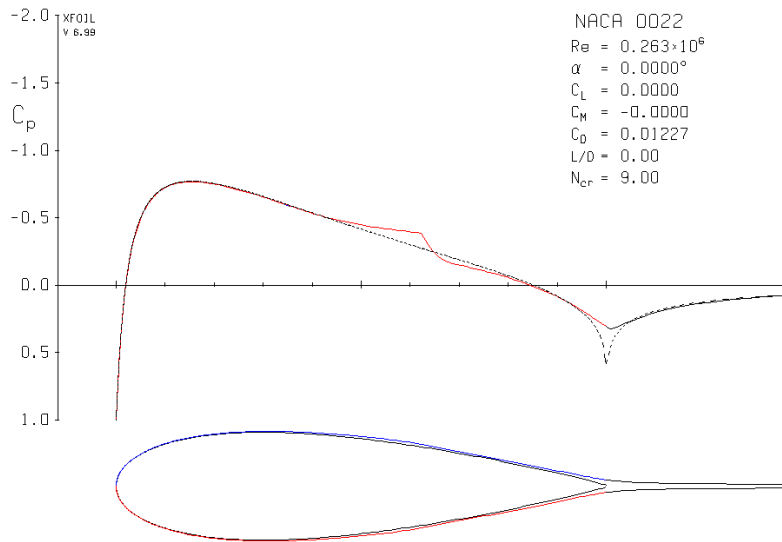
Figure 38: Tensile test results on tested samples

F Drag coefficients for parasitic drag (XFOIL)

XFOIL was used to determine the drag coefficient of the parasitic structure airfoil-shaped elements. The main part of the front struts were likened to NACA0022 at $Re = 263,000$. The bottom part of the front struts, as well as the back strut, were taken as NACA0016 at $Re = 32,000$. The drag coefficients were computed using XFOIL. The results are shown in figure 39. The drag coefficient for the main part of the front struts was found to be 0.01227. It was found to be 0.03778 for the back strut and the bottom parts of the front struts.



(a) Drag coefficient for NACA0016 at Re



(b) Drag coefficient for NACA0016 at Re

Figure 39: XFOIL results for struts drag coefficients